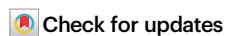


A mutually antagonistic mechanism mediated by RNA m⁶A modification in plant-virus interactions

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RNA-based immunity plays a central role in host defense against pathogens, with both hosts and pathogens continually evolving antagonistic strategies in their ongoing arms race. Although the presence of N⁶-methyl-adenosine (m⁶A) in viruses has been recognized for decades, its functional significance in plant antiviral defenses has only recently been revealed. Moreover, viral counter-strategies targeting m⁶A-mediated defenses remain largely unexplored. Here, we uncover a mutually antagonistic mechanism between m⁶A-mediated antiviral defense and a countermeasure employed by the RNA virus *Cucumber mosaic virus* (CMV). The deposition of m⁶A modification on CMV genomic RNAs was validated through m⁶A antibody-mediated MeRIP and nanopore-based direct RNA sequencing (DRS). During infection, plant m⁶A methyltransferases are translocated to the cytoplasm through their interaction with the viral coat protein (CP), facilitating viral m⁶A deposition. The plant EVOLUTIONARILY CONSERVED C-TERMINAL REGION 8 (ECT8) protein acts as a reader of viral m⁶A, destabilizing viral RNAs and mediating antiviral activity. Conversely, the CMV-2b protein, known as a viral suppressor of RNA silencing (VSR), antagonizes this defense by inhibiting viral m⁶A deposition. This occurs via direct interactions between 2b and the m⁶A methyltransferase components MTB and HAKAI, disrupting the methyltransferase complex's functionality. Furthermore, CMV-2b also downregulates global plant m⁶A levels, leading to the misexpression of defense-related transcripts. Collectively, our findings elucidate a previously unrecognized layer of host-virus interaction in which m⁶A modification serves as a regulatory battleground, positioning m⁶A dynamics as a new frontier in plant-virus coevolution.

Eukaryotic RNA molecules undergo various chemical modifications^{1,2}, and the diversity and abundance of these RNA modifications have significantly expanded over time. Among these, m⁶A is the most prevalent internal modification and was the first to be identified on

mRNA³. The m⁶A modification landscape is regulated by a conserved set of proteins known as “writer”, “reader”, and “erasers”. In the model plant *Arabidopsis thaliana*, the core m⁶A methyltransferase complex (writer) includes mRNA adenosine methylase (MTA), MTB, FK506-

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BINDING PROTEIN 12 KD INTERACTION PROTEIN 37KD (FIP37), VIRILIZER (VIR), HAKAI, HIZ1, and HIZ2^{4–6}. This modification is recognized by a group of YTH domain-containing reader proteins², of which 13 members have been identified in Arabidopsis⁷. Removal of m⁶A marks is achieved by the ALKBH family of demethylases (erasers), with ALKBH9B and ALKBH10B experimentally validated to possess demethylase activity in Arabidopsis^{8–10}.

RNA-based immunity provides plants with a rapid, sequence-specific, and often broad-spectrum defense against both viral and non-viral pathogens^{11,12}. For viral pathogens, double-stranded RNA (dsRNA) can be formed from viral replication intermediates or plant RNA-dependent RNA polymerase (RDR)-mediated amplification of viral RNAs. These virus-derived dsRNAs are processed into small interfering RNAs (siRNAs) by host's RNA interference (RNAi) machinery. These virus-derived siRNAs (vsiRNAs) guide the RNA-induced silencing complex (RISC) to target and degrade viral RNAs, a process known as virus-induced gene silencing (VIGS). This silencing confers post-transcriptional inhibition of RNA viruses or transcriptional inhibition of DNA viruses through RNA-directed DNA methylation (RdDM) pathway. In retaliation, viruses have evolved a suite of viral suppressors of RNA silencing (VSRs) that target various steps in the RNAi pathway¹¹. A prominent example is the CMV-encoded 2b protein, which inhibits siRNA biogenesis and impairs the action of host silencing machinery, thereby diminishing RNAi-mediated antiviral defense¹³.

Beyond antiviral defense, RNAi also contributes to plant immunity against non-viral pathogens. Several microRNAs (miRNAs) have been shown to regulate innate immune receptor homeostasis by targeting their transcripts¹². Importantly, recent studies have revealed that small RNAs can shuttle between host and non-viral pathogens, triggering trans-kingdom RNAi that alters both plant resistance and pathogen virulence. Similar to viral VSRs, some fungal and oomycete pathogens secrete effector proteins that suppress trans-kingdom RNAi, highlighting a convergent evolution of anti-defense strategies^{14–17}. Thus, the arms race between RNAi-based defenses and anti-defense of both viral and non-viral pathogens determines the efficiency of infection. Despite the critical role of RNAi in plant defense, other RNA-based immune mechanisms remain largely unexplored.

Increasing evidence underscores the importance of m⁶A modification in regulating diverse biological processes, including stress responses and pathogen resistance. m⁶A influences RNA metabolism by modulating RNA stability, splicing, export, and translation². Recent findings in both mammals and plants suggest that m⁶A also is an important regulator of antiviral responses. In mammals, m⁶A modification has been detected on the genomes of numerous RNA viruses, and can be recognized by host reader proteins, affecting viral infection^{18–21}. For example, m⁶A facilitates the nuclear transport of human immunodeficiency virus-1 (HIV-1) transcripts¹⁸, while hepatitis C virus (HCV) infection induces the relocalization of m⁶A reader protein YTHDF to lipid droplets (the sites of viral assembly), thereby binding to m⁶A-marked viral RNA and impairing viral infection²¹. Recent findings demonstrate that m⁶A also acts as a significant determinant of plant viral infection⁹. In Arabidopsis, m⁶A modification on alfalfa mosaic virus (AMV) is recognized by the YTH protein ECT2 and removed by ALKBH9B^{9,22}, and loss of ECT2 increases plant susceptibility to AMV, while inactivating ALKBH9B enhances resistance. In rice, viral infection induces global m⁶A accumulation and upregulation of antiviral genes²³. Overexpression of an m⁶A writer from *Solanum lycopersicum*, SlHAKAI, enhances resistance to *Pepino mosaic virus* (PepMV), whereas the virus suppresses SlHAKAI via autophagic degradation mediated by its RNA-dependent RNA polymerase (RdRP)²⁴. In these cases, viral m⁶A modifications enhance plant resistance to viruses. Intriguingly, in some cases, m⁶A modification can also promote viral infection. In wheat, the m⁶A writer TaMTB is hijacked by *Wheat yellow mosaic*

virus (WYMV), facilitating the stabilization of viral RNAs through cytoplasmic m⁶A deposition²⁵. These studies collectively highlight the dual roles of m⁶A in modulating host susceptibility or resistance, depending on the viral context. However, whether and how viruses evolve specific mechanisms to counteract m⁶A-based plant defense remains poorly understood.

In this study, we uncover an antagonistic mechanism employed by a plant virus to suppress m⁶A-based antiviral defense. We validated the deposition of m⁶A modification on CMV genomic RNAs through m⁶A antibody-mediated MeRIP and nanopore-based DRS. During CMV infection, plant m⁶A writers translocate to the cytoplasm via binding to viral CP protein, facilitating m⁶A deposition on viral RNAs. The ECT8 protein specifically binds to these hypermethylated viral RNAs, reducing their stability and thereby restricting viral accumulation. Notably, the CMV-2b protein impedes this process by binding to the methyltransferase components MTB and HAKAI, thereby inhibiting m⁶A deposition. Additionally, CMV-2b broadly suppresses plant m⁶A abundance, leading to the misexpression of many defense-related transcripts. Collectively, our findings reveal a mutually antagonistic mechanism between m⁶A-mediated antiviral immunity and a viral countermeasure, positioning m⁶A dynamics as a pivotal component in the molecular arms race between plant hosts and viruses. To our knowledge, this is the first report of a viral protein directly targeting the m⁶A pathway to suppress host antiviral defense.

Results

m⁶A modification enhances plant resistance to CMV

To elucidate the roles of m⁶A in plant response to CMV infection, we first examined the susceptibility of m⁶A-deficient plants to wild-type CMV. Given that disruption of core m⁶A writers often results in severe developmental abnormalities, we utilized *hakai* mutants, which exhibit approximately a 35% reduction in global m⁶A level in Arabidopsis⁶. Two HAKAI loss-of-function mutants, including a T-DNA insertion mutant (*hakai-1*) and a CRISPR/Cas9-edited mutant (*hakai-c*, Supplementary Fig. 1a), along with two complementary lines of the *hakai-1* mutant (*HAKAIcom*, Supplementary Fig. 1c), were examined. The reduction of m⁶A levels in the *hakai* mutants was validated by m⁶A dot blotting (Supplementary Fig. 1b). Surprisingly, both *hakai* mutants exhibited disease symptoms (Fig. 1a) and viral RNA accumulation comparable to wild-type Col-0 and *HAKAIcom* plants (Fig. 1b, c, and Supplementary Fig. 2). To further explore the involvement of m⁶A in CMV resistance, we analyzed two m⁶A demethylase mutants, *alkbh9b* and *alkbh10b-c* (Supplementary Fig. 1a and b), using the same infection assay. Consistent with the *hakai* mutant results, both demethylase mutants also displayed no significant difference in disease symptoms and viral titers compared to wild-type plants (Supplementary Fig. 3).

CMV encodes the well-characterized VSR, the 2b protein, which is essential for CMV virulence¹³. We hypothesized that the presence of the 2b protein might obscure the role of m⁶A in plant antiviral defense. To test this, we employed a 2b-deficient virus (CMV-Δ2b), which ablates 2b expression without affecting 2a translation²⁶. Intriguingly, both *hakai* mutants exhibited significantly increased susceptibility to CMV-Δ2b compared to Col-0 (Fig. 1a), and higher viral RNA accumulation in the systemic leaves (Fig. 1b, c, and Supplementary Fig. 2). The *hakai* mutants infected with CMV-Δ2b virus showed more obvious growth inhibition than the wild-type Col-0 plants. Importantly, complementation with genomic *HAKAI* sequence fully restored resistance in the *hakai* mutant background (Fig. 1a–c and Supplementary Fig. 2). Conversely, both demethylase mutants showed enhanced resistance to CMV-Δ2b, as evidenced by reduced symptom severity and decreased viral RNA accumulation relative to Col-0 plants (Supplementary Fig. 3). Collectively, these data suggest that m⁶A modification enhances plant antiviral defense against CMV, and that this defense is actively suppressed by the 2b protein.

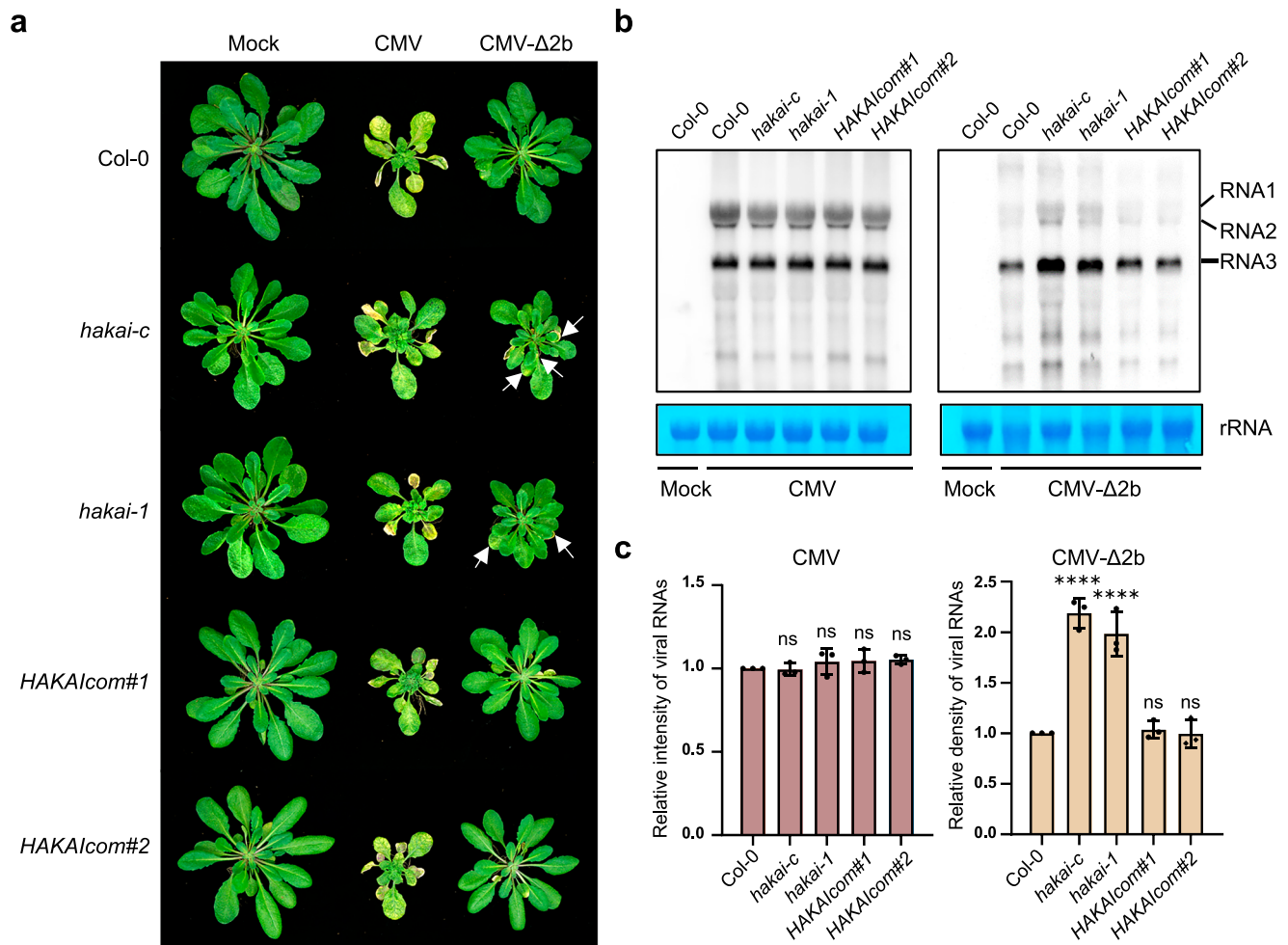


Fig. 1 | The *hakai* mutants display increased susceptibility to CMV-Δ2b but not to wild-type CMV. **a** Disease symptoms observed in Col-0, *hakai* mutants, and *HAKA1com* lines at 14 days post-infection (dpi) with either wild-type CMV or CMV-Δ2b. White arrows indicate the yellowed leaves caused by CMV-Δ2b infection. **b** RNA blotting analysis of viral RNA accumulation in the systemic leaves from the plants shown in (a). Methyl blue staining serves as a loading control of total RNA. Biotin-labeled PCR fragments amplified from the conserved 3'UTR regions of CMV

RNA1, RNA2, and RNA3 were used as blotting probes. **c** Column diagrams showing the relative intensity of viral RNAs calculated based on three biological independent replicates (Fig. 1b and Supplementary Fig. 2). Data are the means $\pm$ SD ($n = 3$ biological independent replicates). Significance values were calculated with unpaired two-sided *t*-test (**** $p < 0.0001$; ns, no significant). Source data are provided as a Source Data file.

m⁶A modification-mediated antiviral defense functions independently of the RNAi pathway

Given that the CMV-2b protein is a well-characterized VSR, we investigated whether the increased sensitivity of *hakai* mutants to CMV-Δ2b could be attributed to a secondary effect caused by the release of 2b-mediated inhibition of antiviral RNAi defense. To test this possibility, we examined the accumulation of virus-derived siRNAs in Col-0 and *hakai* mutants upon infection with either CMV or CMV-Δ2b. As a positive control, we included the *dcl2/4* double mutant, which lack the two major Dicer-like proteins responsible for processing 21–22 nt vsiRNAs^{27–31}. As expected, small RNA blotting analysis revealed a significant reduction in 21–22 nt vsiRNAs and a concomitant increase in 24 nt vsiRNAs in both CMV- and CMV-Δ2b-infected *dcl2/4* mutants (Supplementary Fig. 4), consistent with the known role of DCL2 and DCL4 in antiviral RNAi. By contrast, *hakai* mutants exhibited vsiRNA profiles nearly identical to those of Col-0 plants under both infection conditions. These findings suggest that m⁶A modification-mediated antiviral defense does not alter the accumulation of vsiRNAs, and therefore functions independently of the canonical RNAi antiviral pathway.

Identification of m⁶A modification sites on viral RNAs

To determine whether CMV genomic RNAs are directly modified by m⁶A, we first performed methylated RNA immunoprecipitation sequencing (MeRIP-seq) assay on total RNA extracted from Col-0 plants that were mock-treated or infected with CMV or CMV-Δ2b. For the analysis of MeRIP-seq data, the IP sample represents mRNA fragments immunoprecipitated using m⁶A-specific antibody, while the input sample corresponds to total mRNA without enrichment, serving as a background control. The MeRIP-seq analysis revealed three discrete m⁶A peaks on RNA1 and RNA2 of both CMV and CMV-Δ2b genomes (Fig. 2a and Supplementary Data 1). Although a high enrichment of both IP and input reads was observed on the RNA3, the lack of a significant difference in read abundance between the two data indicates that this region should not be considered an m⁶A modification peak.

A recent study has underscored a key limitation of m⁶A antibody-dependent techniques, which are prone to a high rate of false positives³². To overcome this limitation and validate the presence of m⁶A modification on viral RNAs, we employed an m⁶A antibody-free method, nanopore direct RNA sequencing (DRS). This method enables

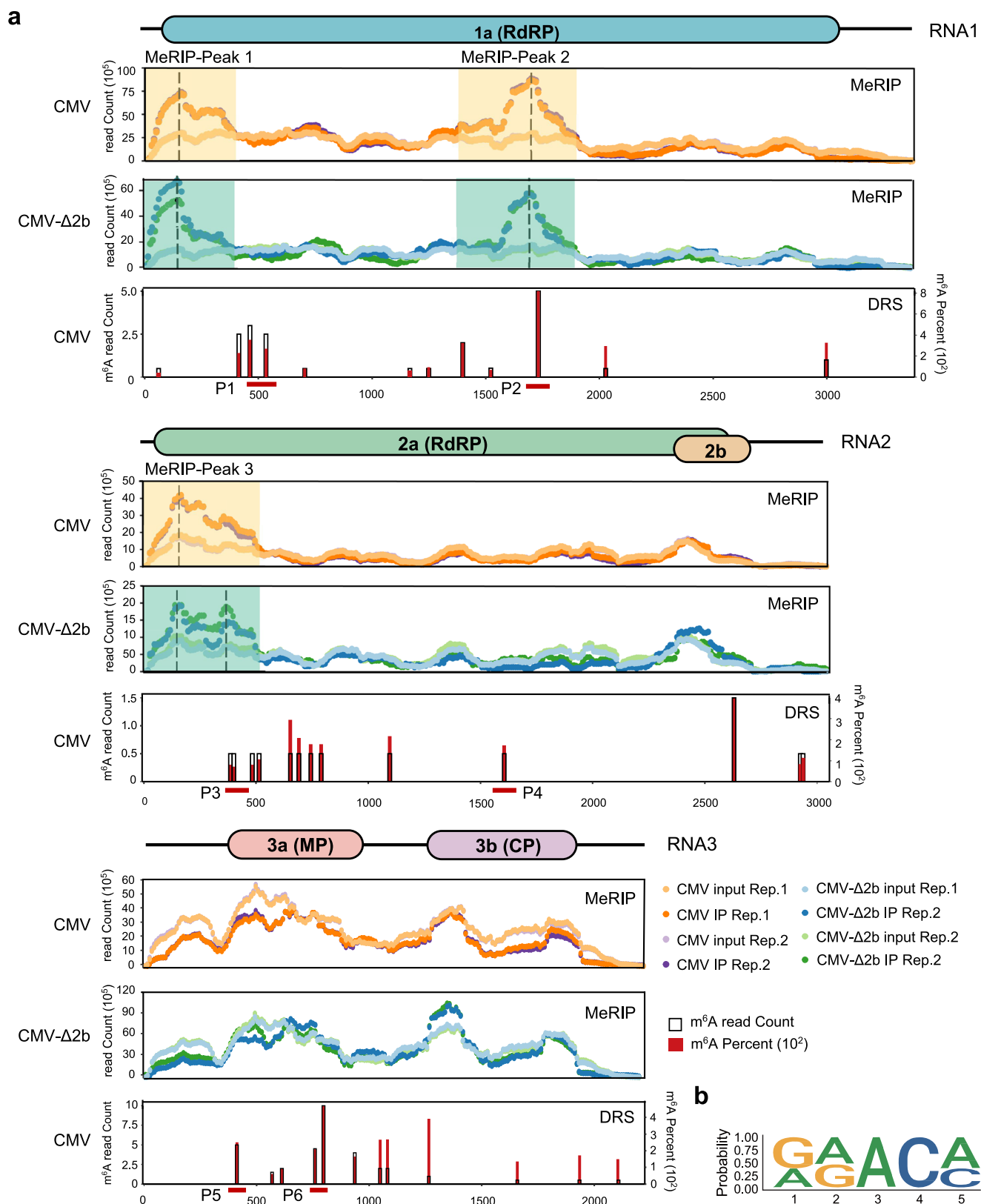


Fig. 2 | Identification of m⁶A modification and regions on CMV genomic RNAs.
a Snapshots of MeRIP-seq showing reads distribution across CMV and CMV-Δ2b viral RNAs. DRS snapshots showing m⁶A sites distribution across CMV viral RNAs. Identified m⁶A peaks and precise sites are indicated. Red lines

indicate the positions of primer pairs used for quantitative PCR analysis.
b m⁶A Motif enrichment for DRS. Motif finding by HOMER using all identified m⁶A sites on CMV genomic RNAs. Source data are provided as a Source Data file.

the detection of m⁶A modifications at single-nucleotide resolution. The DRS assay was performed using CMV-infected Arabidopsis Col-0 plants. By aligning the sequencing data to the CMV reference genome, we identified a total of 37 m⁶A modification sites distributed across all three CMV genomic RNAs (Fig. 2a and Supplementary Data 2), with 12, 13, and 12 m⁶A sites mapped to RNA1, RNA2, and RNA3, respectively. Notably, most of the m⁶A sites on RNA1 and RNA2 corresponded to regions previously identified as peaks (peak 1–3) and their flanking regions by the MeRIP-seq method. In contrast, MeRIP-seq failed to detect any m⁶A peaks on RNA3. These findings suggest that while MeRIP-seq accurately identifies many m⁶A-modified regions, it is also associated with false positives and may overlook certain modifications, particularly in specific genomic contexts. We speculate that this may be caused by variable antibody affinity or peak-calling stringency. Importantly, the DRS data provide robust and direct evidence for the presence of m⁶A modifications across all three CMV genomic RNAs. Additionally, a conserved “RRACM” motif was identified among the DRS-mapped m⁶A sites, further supporting the specificity of the modifications (Fig. 2b).

Next, to validate the in vivo deposition of m⁶A on viral RNAs, m⁶A immunoprecipitation (m⁶A-IP) was performed in Col-0 and the *hakai-c* mutant under CMV infection. Quantitative PCR was used to assess m⁶A enrichment surrounding six m⁶A sites identified by DRS. The m⁶A-IP results indicated a significant reduction in m⁶A enrichment at all six DRS sites in the *hakai-c* mutant compared to Col-0. In contrast, the IgG-IP control displayed minimal background signal and no significant difference between Col-0 and the *hakai-c* mutant (Fig. 3a). These findings support the role of m⁶A methyltransferases in recognizing and depositing m⁶A on viral RNAs.

m⁶A modification disrupts viral RNA stability

Next, to assess the functional consequence of m⁶A modification on viral RNA stability, we investigate the decay of viral genomic RNAs in Col-0 and *hakai-c* mutants following cordycepin treatment, as previously described³³. The results revealed that all three CMV genomic RNAs exhibited remarkably enhanced stability in the *hakai-c* mutant compared to Col-0 plants (Fig. 3b), suggesting that m⁶A modification negatively regulates viral RNA stability. To further confirm this observation, RNA stability assays were also conducted in the m⁶A demethylase mutant *alkb10b-c*. In contrast to *hakai-c*, *alkb10b-c* mutant showed substantially reduced stability compared to Col-0 plant (Fig. 3c). Taken together, the in vivo data strongly indicate that the presence of m⁶A modification compromises the stability of viral RNAs.

CMV infection induces the cytoplasmic translocation of m⁶A writers

It is well established that m⁶A methyltransferases primarily catalyze m⁶A modification on mRNA within the nucleus³⁴. However, several plant RNA viruses replicate within the cytoplasm or plastids, raising questions about how the m⁶A writer complex accesses viral RNA substrates at the subcellular level. To address this, we compared the subcellular localization patterns of m⁶A writers under mock versus CMV infection conditions by transiently expressing these proteins in *N. benthamiana* leaves. As shown in Fig. 4a, m⁶A writers such as MTB, HAKAI, and FIP37 were primarily localized to the nucleus under mock conditions, although HAKAI exhibited faint cytoplasmic fluorescence. Intriguingly, upon CMV infection, a remarkable translocation of m⁶A writers to the cytoplasm was observed, particularly for MTB and FIP37. In contrast, the localization of free GFP remained unchanged between mock and CMV-infected samples, serving as a control. These results suggest that CMV infection induces the cytoplasmic translocation of m⁶A writers, potentially enabling direct m⁶A deposition on viral RNA.

To further confirm this observation, we investigated the in vivo subcellular localization of MTB and FIP37, which predominantly localized to the nucleus (Fig. 4a) under mock and CMV infection conditions. The *MTBpro:MTB-GFP* and *FIP37pro:FIP37-GFP* transgenic Arabidopsis plants driven by their native promoter were employed. As depicted in Fig. 4b and Supplementary Fig. 5, the cytoplasmic accumulation of MTB and FIP37 proteins was dramatically increased upon CMV infection, while the nuclear accumulation was greatly reduced. Therefore, these results strongly support our conclusion that CMV infection induces the cytoplasmic translocation of m⁶A writers.

m⁶A methyltransferases binds to CMV-CP protein to promote viral m⁶A deposition in the cytoplasm

To elucidate the molecular mechanism underlying the cytoplasmic translocation of m⁶A writers, we performed protein interaction analysis between viral proteins and m⁶A methyltransferases. Yeast two-hybrid analysis (Y2H) revealed a strong interaction between the CMV-CP protein and the core methyltransferase MTB (Fig. 4c and Supplementary Fig. 6a). To validate the in vivo interaction of MTB and CMV-CP, co-IP assay was performed in transgenic *MTBpro:MTB-GFP* plants under CMV infection condition. CP protein was detected using endogenous anti-CP antibody. As expected, CP protein can be immunoprecipitated by MTB in vivo (Fig. 4d). These results suggest that m⁶A writers may be recruited to viral RNA via direct binding to CMV-CP protein in the cytoplasm. Supporting this hypothesis, subcellular localization analysis showed that the CP protein, which predominantly localizes to the nucleus under mock conditions, relocates to the cytoplasm upon CMV infection (Fig. 4e). The translocation was further validated through a bimolecular fluorescence complementation (BiFC) assay in *N. benthamiana* leaves. Under mock conditions, CP-MTB interaction primarily occurred in the nucleus. However, following CMV infection, interaction signals were primarily cytoplasmic (Fig. 4f). Collectively, these results demonstrate that m⁶A writers are redirected from the nucleus to the cytoplasm via direct interaction between CP and MTB, ultimately facilitating m⁶A deposition on viral RNA.

ECT8 functions as a reader of viral m⁶A modification

In plants, YTH domain-containing proteins serve as m⁶A readers³⁵. To identify candidate readers that recognize viral m⁶A modifications, we first examined the transcriptional response of m⁶A reader genes upon CMV infection. RNA-seq analysis revealed that several m⁶A reader genes were upregulated, with *ECT7*, *ECT8*, and *ECT10* showing the most pronounced induction (Fig. 5a). Notably, mutants lacking these genes displayed significantly heightened susceptibility to CMV-Δ2b infection compared to Col-0 plants (Fig. 5b). Consistently, these mutants accumulated significantly higher levels of viral RNA (Fig. 5c, d, and Supplementary Fig. 7a). In contrast, no significant difference in disease symptoms or viral RNA accumulation was observed between *ect7*, *ect8*, *ect10* mutants and Col-0 under CMV infection. Furthermore, all other *ect* mutants exhibited no significant difference from Col-0 under both CMV and CMV-Δ2b infection conditions (Supplementary Fig. 7b–d).

Recent studies have demonstrated that ECT8 promotes the decay of m⁶A-modified mRNA by interacting with the decapping protein DCP5^{36,37}. Considering the increased susceptibility of the *ect8* mutant to CMV-Δ2b (Fig. 5b–d) and the compromised stability of m⁶A-marked viral RNA (Fig. 3b and c), we hypothesize that ECT8 may act as a reader of viral m⁶A modification. To confirm this hypothesis, we examined the binding capacity of ECT8 to m⁶A-modified viral RNA using an electrophoretic mobility shift assay (EMSA). Two viral RNA probes corresponding to the DRS m⁶A sites of viral RNA1 and RNA2 were synthesized with or without m⁶A modification. These probes were incubated with purified recombinant GST-ECT8 protein expressed in *E. coli*, with GST serving as a control protein (Supplementary Fig. 8a). As shown in Fig. 5e, GST-ECT8 exhibited robust and specific binding to

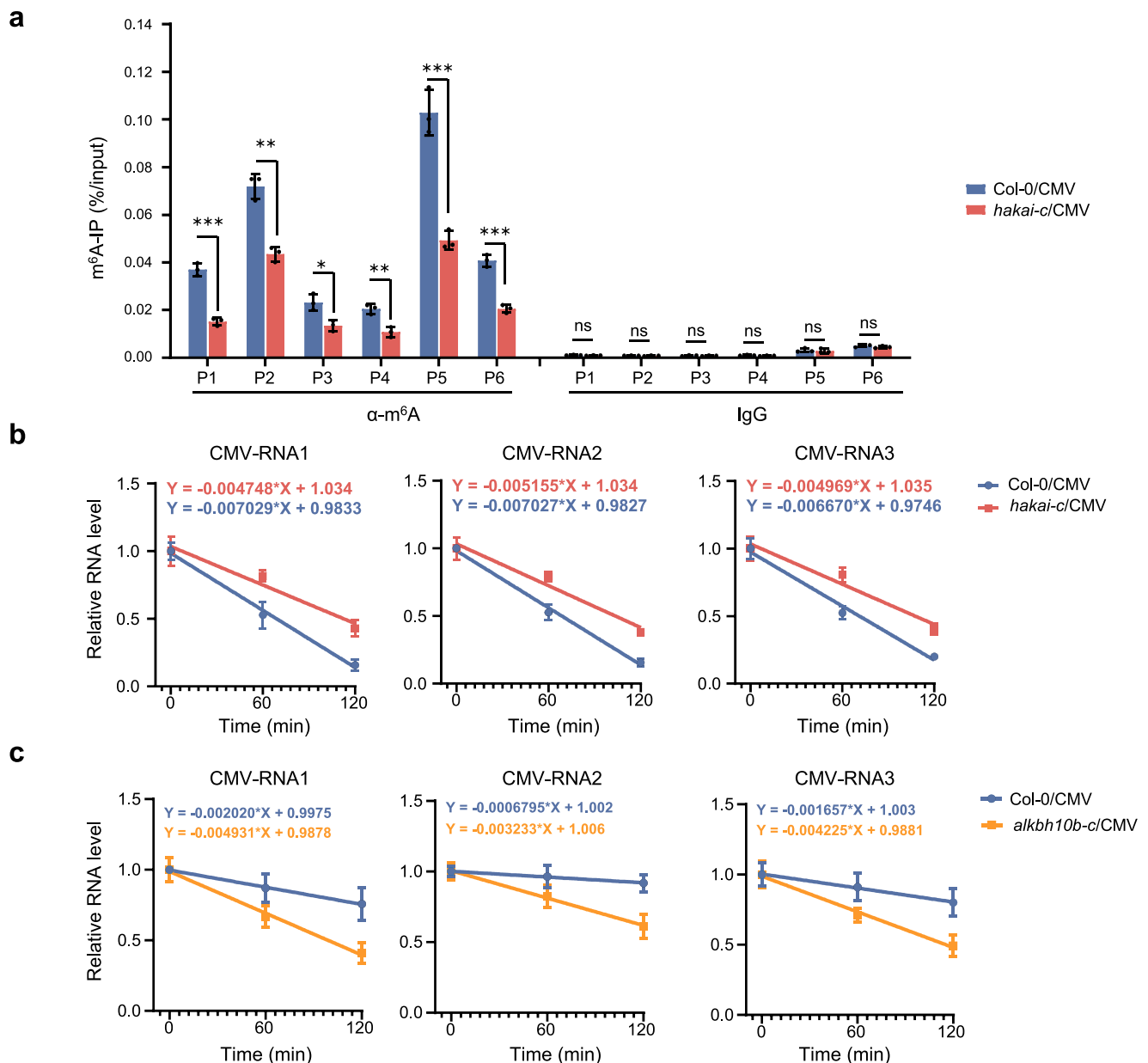


Fig. 3 | m⁶A modification disrupts viral RNA stability. **a** m⁶A IP-qPCR results showing the relative m⁶A levels on different regions of three viral RNAs corresponding to six DRS sites in Col-0 and *hakai-c* mutant under CMV infection. IP assays were performed using anti-m⁶A antibody. IgG IP was performed as negative controls. Data are the means $\pm$ SD ($n = 3$ biological independent replicates). Significance values were calculated with unpaired two-sided *t*-test (**** $p < 0.001$, *** $p < 0.01$, ** $p < 0.05$). P1, $p = 0.0003$; P2, $p = 0.0012$; P3, $p = 0.0143$; P4, $p = 0.0046$;

P5, $p = 0.0009$; P6, $p = 0.0004$. **b, c** Line charts showing the stability of three viral RNAs in Col-0 and *hakai-c* mutant (**b**) or Col-0 and *alkbh10b-c* mutant (**c**). Data are the means $\pm$ SD ($n = 3$ biological independent replicates). One representative result of three independent biological replicates was shown. Samples were collected at 7 days post-CMV infection. *AT4G33380* serves as an internal control to calculate the relative RNA level. Source data are provided as a Source Data file.

the m⁶A-modified probes, with negligible affinity for the unmodified probes (Fig. 5e). The binding intensity increased in a concentration-dependent manner. In contrast, GST protein showed no binding to both m⁶A-modified and unmodified probes (Supplementary Fig. 8b).

To validate the *in vivo* binding of ECT8 to m⁶A-modified viral RNA, we performed RNA immunoprecipitation followed by qPCR (RIP-qPCR) using Col-0 and transgenic *ECT8pro::YFP-ECT8* Arabidopsis plants under CMV infection to investigate the *in vivo* association between ECT8 and viral RNAs. As shown in Fig. 5f, viral RNA fragments encompassing the six m⁶A sites identified by DRS were significantly enriched in RNAs immunoprecipitated from *ECT8pro::YFP-ECT8* plants compared to those from Col-0. In contrast, the negative control *ACT2* locus showed minimal enrichment, with no significant difference between the Col-0 and

transgenic lines. The *in vitro* and *in vivo* evidence strongly supports the conclusion that ECT8 functions as a bona fide m⁶A reader that specifically recognizes m⁶A-modified viral RNAs *in vivo*.

CMV-2b protein impedes m⁶A deposition on viral RNAs

The enhanced susceptibility of m⁶A-deficient mutants to CMV-Δ2b prompts us to hypothesize that 2b protein may inhibit m⁶A deposition on viral RNAs. To test this hypothesis, we first analyzed m⁶A enrichment in CMV and CMV-Δ2b using MeRIP-seq data. The results indicated that CMV-Δ2b exhibited significantly higher m⁶A level at the peak 1 region of RNA1 and the peak 3 region of RNA2 compared to CMV (Figs. 2a and 6a), while no substantial difference was observed at the peak 2 region of RNA1.

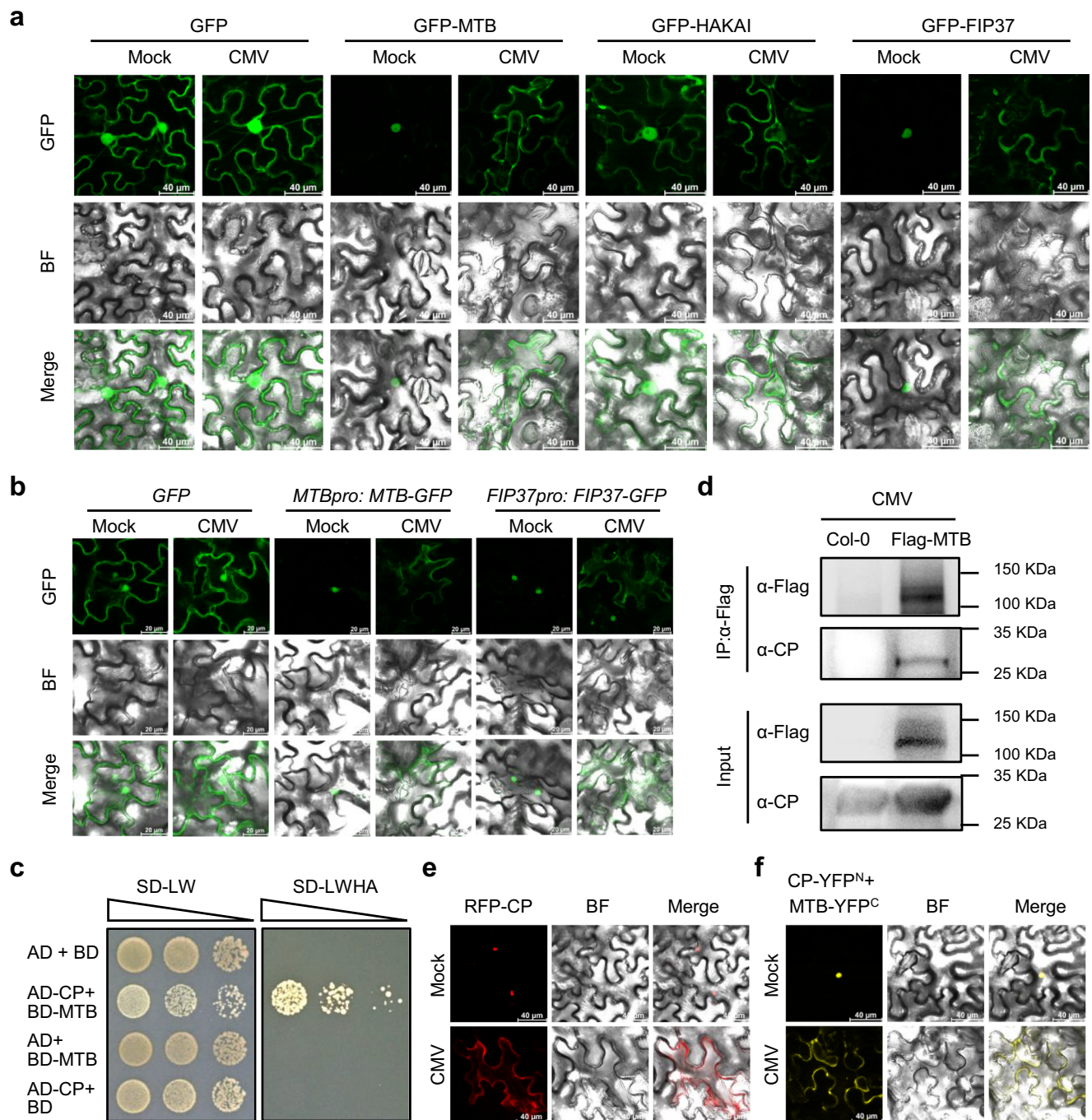


Fig. 4 | CMV infection induces cytoplasmic translocation of m⁶A writers through CP-MTB interaction. **a** CMV infection promotes cytoplasmic translocation of m⁶A writers. MTB, HAKAI, and FIP37 were transiently expressed in *N. benthamiana* leaves under mock or CMV-infected conditions. Free GFP serves as a control. Fluorescence was imaged at 2 dpi. BF, bright field. **b** CMV infection induces the in vivo cytoplasmic translocation of MTB and FIP37. The localization of MTB and FIP37 was investigated in the *MTBpro:MTB-GFP* and *FIP37pro:FIP37-GFP* transgenic Arabidopsis plants under mock or CMV-infected conditions. Transgenic *35S:GFP* plants serve as negative control. Fluorescence was imaged one week after CMV infection. **c** Y2H assay showing a direct interaction between CMV-CP and MTB. **d** Co-IP assay showing the in vivo association of MTB and CP. Co-IP assays

were performed in Col-0 and the *Flag-MTB* transgenic Arabidopsis plants under the condition of CMV infection using anti-Flag antibody. MTB and viral CP proteins were detected using anti-Flag and anti-CP antibodies, respectively. **e** Subcellular localization of CP protein fused with RFP under mock and CMV infection. RFP-CP protein was transiently expressed in mock or CMV pre-infected *N. benthamiana* leaves. RFP fluorescence was detected at 2 dpi. **f** BiFC assay in *N. benthamiana* leaves showing that CP-MTB interactions shift from predominantly nuclear (mock) to cytoplasmic (CMV) localization. Y^N and Y^C represent the N- and C-terminal halves of YFP, respectively. YFP fluorescence was detected at 2 dpi. Source data are provided as a Source Data file.

To validate this observation, viral particles were isolated from plants infected with either CMV or CMV-Δ2b. Viral genomic RNAs were extracted and subjected to m⁶A dot blot analysis using an m⁶A-specific antibody. As expected, CMV-Δ2b particles exhibited

approximately double the m⁶A levels observed in CMV particles (Fig. 6b).

To more accurately quantify m⁶A levels in CMV and CMV-Δ2b, we performed liquid chromatography and tandem mass spectrometry

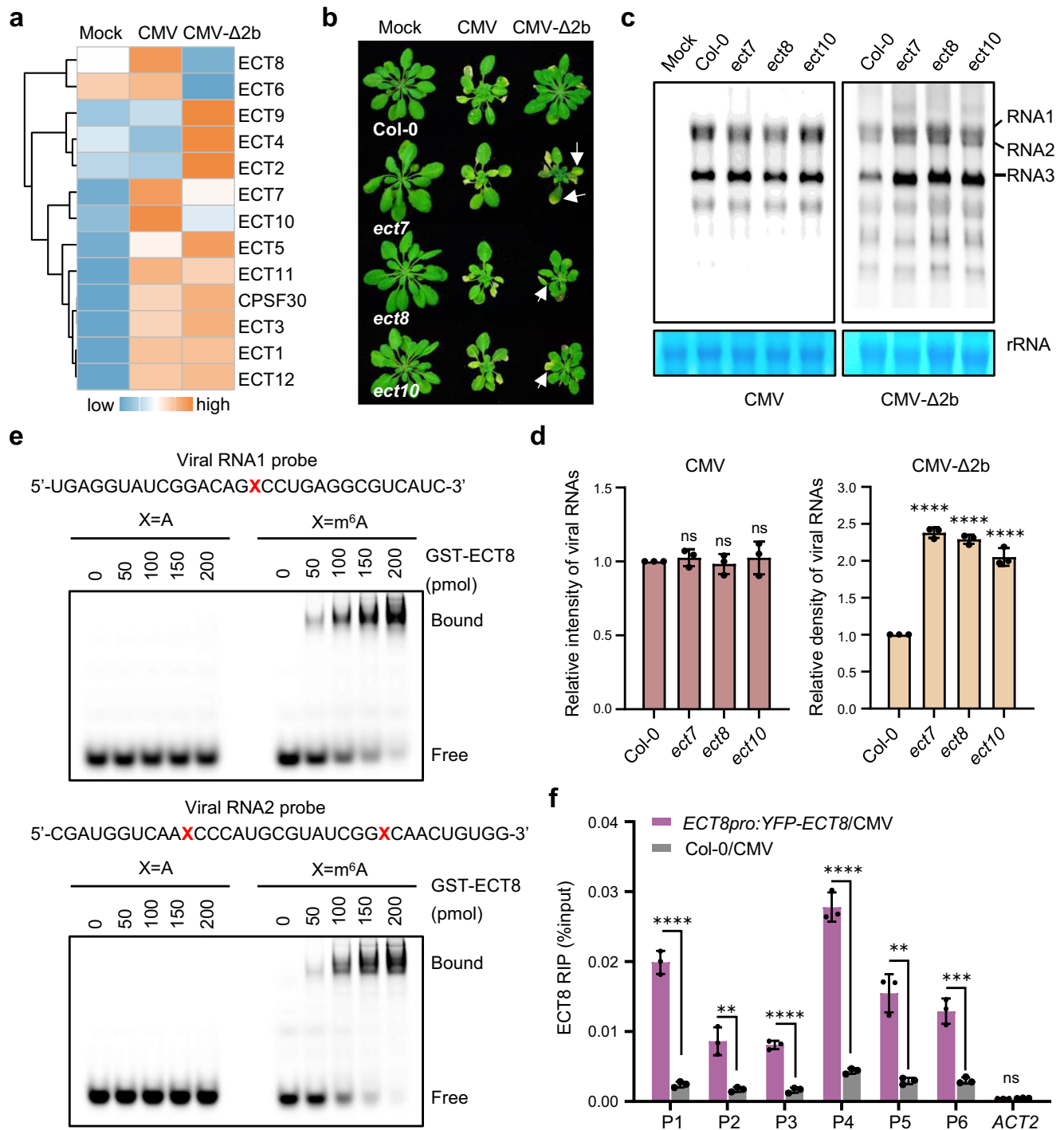
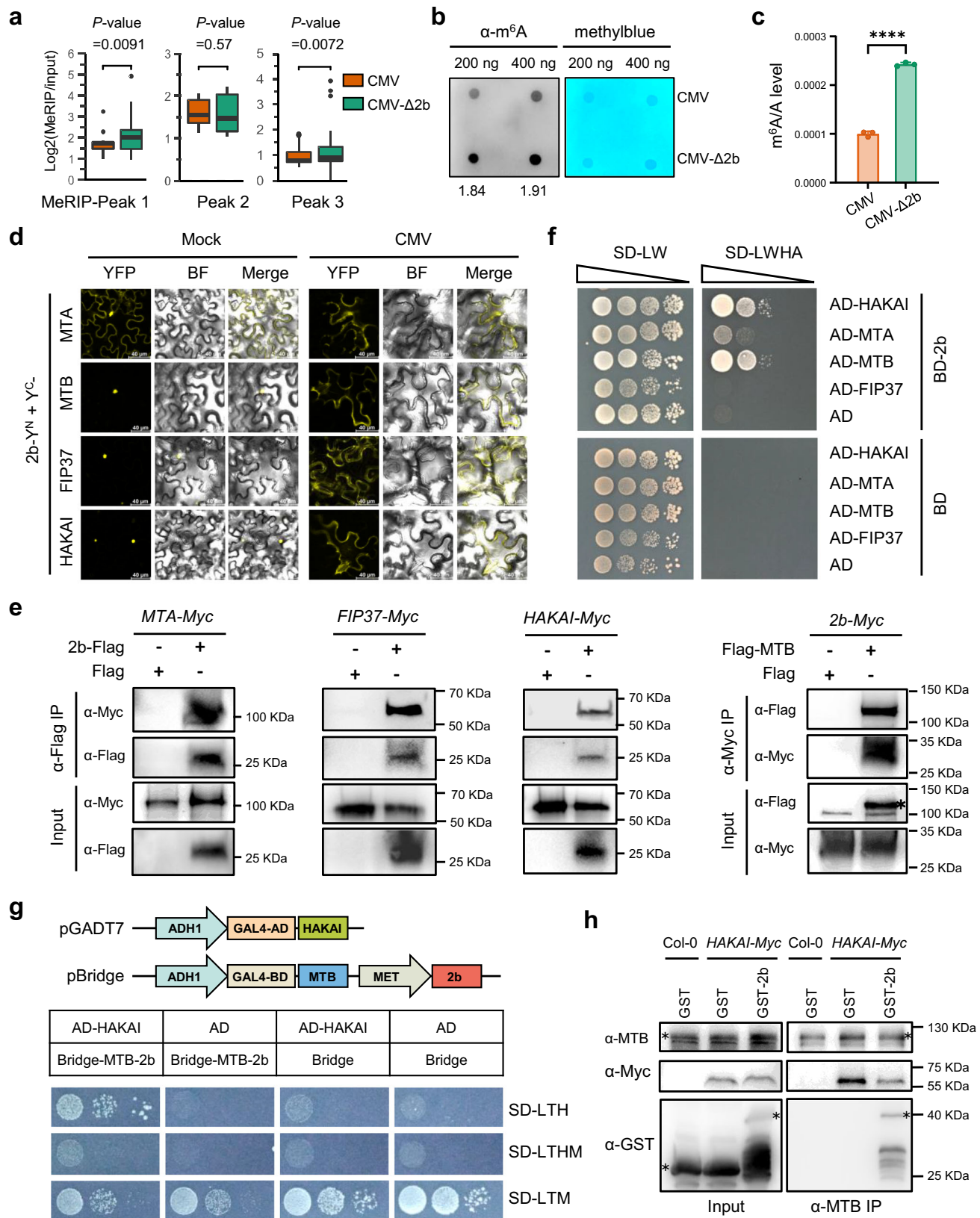


Fig. 5 | ECT8 functions as a reader of viral m⁶A modification. **a** Heatmap showing the relative expression of m⁶A reader genes in mock-treated and CMV-infected Col-0 plants. **b** Disease symptoms in Col-0, *ect7*, *ect8*, and *ect10* mutants infected with CMV or CMV-Δ2b. White arrows indicate the yellowed leaves caused by CMV-Δ2b infection. Photographs were taken at 14 dpi. **c**, **d** RNA blotting results showing the accumulation of viral RNA in the systemic leaves of CMV- and CMV-Δ2b-infected plants (c). The relative intensity of viral RNAs (d) was quantified based on three independent replicates (c and Supplementary Fig. 7a). Data are the means ± SD ($n = 3$ biological independent replicates). Significance values were calculated with unpaired two-sided t-test (**** $p < 0.0001$; ns, no significant). **e** EMSA assay

showing a high binding affinity of GST-ECT8 to the m⁶A-modified viral RNA probes. Sequence of viral RNA1 and RNA2 probes are shown in the upper panel. **f** ECT8 displays in vivo binding affinity to m⁶A-modified viral RNAs. ECT8 RIP assay was performed in Col-0 and the transgenic *ECT8pro::YFP-ECT8* Arabidopsis plants under CMV infection. The relative enrichment of viral RNA bound by ECT8 at different regions was quantified. *ACT2* locus serves as negative control. Data are the means ± SD ($n = 3$ biological independent replicates). Significance values were calculated with two-sided t-test (**** $p < 0.0001$, **** $p < 0.001$, *** $p < 0.01$). P1, $p < 0.0001$; P2, $p = 0.0041$; P3, $p < 0.0001$; P4, $p < 0.0001$; P5, $p = 0.0014$; P6, $p = 0.0008$. Source data are provided as a Source Data file.



(LC-MS/MS) analysis of viral genomic RNAs extracted from CMV and CMV-Δ2b particles. As shown in Fig. 6c and Supplementary Data 3, the m⁶A/A ratio is approximately 0.0001 in CMV, while this value is significantly higher in CMV-Δ2b, reaching approximately 0.00025. Collectively, these findings strongly support the notion that the CMV-2b protein impedes the deposition of m⁶A on viral RNAs.

2b protein interacts with MTB and HAKAI to disrupt m⁶A writer complex functionality

The inhibitory effect of 2b on m⁶A deposition inspires us to explore its potential interference with the m⁶A writer complex. To this end, we examined the potential interactions between 2b and the m⁶A writers. 2b protein is predominantly localizes in the nucleus, and a certain

Fig. 6 | CMV-2b impairs m⁶A deposition through interactions with MTB and HAKAI. **a** Boxplot of m⁶A enrichment on the genomic RNAs of CMV and CMV-Δ2b viruses from infected *Arabidopsis thaliana*. Enrichment was quantified using MeRIP-seq data. Data are the means ± SD ($n = 3$ biological independent replicates). Each dot represents the Log2(MeRIP/input) for an individual m⁶A peak. Significance values were calculated with two-sided Student's t-test (peak 1, $p = 0.0091$; peak 2, $p = 0.57$; peak 3, $p = 0.0072$). The boxplot displays the median (center line), the first (25th percentile) and third (75th percentile) quartiles as the box bounds, and whiskers extending to the minimum and maximum values within 1.5 times the interquartile range. **b** Dot blot analysis of m⁶A levels in the RNAs isolated from CMV and CMV-Δ2b particles. Methylblue staining indicates the loading control of total viral RNAs. **c** LC-MS/MS quantification of the m⁶A/A ratio in the genomic RNAs of CMV and CMV-Δ2b viruses. Data are the means ± SD ($n = 3$ biological independent replicates). Significance values were calculated with two-sided t-test (*****) $p < 0.0001$. **d** BiFC showing the subcellular localization of CMV-2b interactions

with m⁶A writers in mock (left) and CMV-infected (right) *N. benthamiana* leaves. YFP fluorescence was measured at 2 dpi. The negative controls are shown in the Supplementary Fig. 9b. **e** Co-IP showing the in vivo association of CMV-2b with m⁶A writers. 2b-Flag or 2b-Myc and m⁶A writers were transiently co-expressed in *N. benthamiana* leaves, and α-Flag antibody immunoprecipitation was performed. The Flag tag or Myc tag co-expressed with m⁶A writers served as a negative control. Asterisk marks Flag-MTB protein. **f** Y2H assay showing direct interactions between CMV-2b and HAKAI and MTB. **g** A Y3H assay indicates that CMV-2b inhibits the interaction between HAKAI and MTB. Schematic diagram of constructs shown in upper panel. **h** Competitive Co-IP showing that 2b disrupts the MTB-HAKAI interaction. Co-IP was performed in Col-0 and 35S:HAKAI-Myc *Arabidopsis* using anti-MTB antibody with GST or GST-2b. MTB, HAKAI-Myc, GST, and GST-2b were detected with the indicated antibodies. Asterisks mark MTB, GST, and GST-2b proteins. Source data are provided as a Source Data file.

amount is also accumulated in the cytoplasm (Supplementary Fig. 9a). CMV infection increases the cytoplasmic accumulation of 2b protein, but it still maintains a large amount of accumulation in the nucleus (Supplementary Fig. 9a). These patterns of 2b are consistent with the previous documents^{33,38–40}. The BiFC assay demonstrated that 2b associates with all four tested subunits of the m⁶A writer complex, including MTA, MTB, HAKAI, and FIP37, in the nucleus of *N. benthamiana* cells under mock conditions (Fig. 6d and Supplementary Fig. 9b). These interactions were further confirmed by co-IP assays via transient co-expression in *N. benthamiana* leaves (Fig. 6e). Importantly, the interaction fluorescence between 2b and these m⁶A writers was mainly detected in the cytoplasm (Fig. 6d), implying a substantial translocation from nucleus to cytoplasm. Y2H assays further revealed that 2b directly interacts with HAKAI and MTB, and has a mild interaction with MTA. No interaction between 2b and FIP37 was observed (Fig. 6f and Supplementary Fig. 6b).

Given that MTB and HAKAI directly interact (Supplementary Fig. 10)⁶ and are both known targets of the CMV-encoded 2b protein, we hypothesized that 2b may interfere with m⁶A deposition by disrupting the MTB–HAKAI interaction. To test this hypothesis, we performed a Y3H assay to determine whether 2b interferes with the MTB–HAKAI interaction. In this system, 2b expression is driven by the *MET* promoter and activated upon methionine withdrawal. The Y3H results clearly demonstrated that 2b expression significantly impaired the interaction between HAKAI and MTB (Fig. 6g).

To further validate the inhibition, we compared the in vivo interaction between MTB and HAKAI in the presence of purified GST-2b protein or the control GST protein in Col-0 and transgenic *HAKAI-Myc Arabidopsis* plants via anti-MTB IP assays. As shown in Fig. 6h, HAKAI-Myc protein was detected in the IP samples containing GST or GST-2b, supporting the known in vivo interaction between MTB and HAKAI. Importantly, the amount of immunoprecipitated HAKAI-Myc protein by MTB was greatly reduced in the presence of GST-2b compared to GST, strongly suggesting that GST-2b impairs the interaction between MTB and HAKAI. Together, these results support the conclusion that the 2b protein disrupts the functionality of the m⁶A writer complex by competitively binding to its core components, thereby impairing m⁶A deposition.

CMV-2b inhibits plant m⁶A deposition

Next, we asked whether CMV-2b has a substantial impact on plant m⁶A homeostasis during viral infection. To confirm this, we profiled m⁶A landscapes in mock-treated, CMV-, and CMV-Δ2b-infected Col-0 plants. We identified 14,371, 15,757, and 16,484 m⁶A peaks, respectively (Fig. 7a and Supplementary Data 4), predominantly located in 3' untranslated regions (3'UTR), indicating that viral infection did not significantly alter the distribution pattern of m⁶A in plants. The conserved “UGUAA” motif was consistently enriched across all conditions⁴¹, whereas an additional “GAACC” motif appeared

specifically in CMV-infected plants (Supplementary Fig. 11). Compared to mock plants, CMV infection resulted in 181 hypermethylated and 991 hypomethylated m⁶A peaks, while CMV-Δ2b induced only 37 hyper- and 186 hypomethylated peaks (Supplementary Data 4). Direct comparison between CMV and CMV-Δ2b revealed 56 hyper and 85 hypo-m⁶A peaks (Fig. 7a). Notably, CMV infection caused a broader decrease in m⁶A levels compared to CMV-Δ2b (Fig. 6b, c). To validate this observation, we overexpressed 2b protein in Col-0, and the transgenic *2b-Myc* plants were subjected to m⁶A dot blot analysis (Fig. 7d). A significant reduction in m⁶A levels in *2b-Myc* plants was observed compared to wild-type plants. Consistent with the dot blot result, LC-MS/MS analysis demonstrated that 2b expression significantly decreased m⁶A level compared to Col-0 (Fig. 7e and Supplementary Data 3). These results suggest that CMV-2b has a substantial impact on plant m⁶A homeostasis, primarily leading to a downregulation of plant m⁶A levels.

CMV-2b alters the expression of m⁶A-marked defense-related genes

Next, to assess how 2b-mediated inhibition of plant m⁶A deposition on the plant transcriptome, a conjoint analysis of MeRIP-seq and RNA-seq data was conducted. For CMV-induced m⁶A-hypomethylated genes, 223 genes were significantly upregulated, while only 19 genes were downregulated compared to mock plants (Fig. 7f, left panel). In contrast, CMV-induced m⁶A-hypermethylated genes showed minimal transcriptional changes. Notably, CMV-Δ2b had a negligible impact on both m⁶A methylation and gene expression compared to mock (Fig. 7f, middle panel), further supporting the role of 2b in m⁶A inhibition. These results are consistent with previous results that mock and CMV-Δ2b-infected plants exhibited similar, yet higher, m⁶A levels compared to CMV-infected plants, with most genes being m⁶A hypomethylated (Fig. 7b, c). Collectively, these data strongly suggest that CMV infection-mediated downregulation of plant m⁶A levels, primarily driven by the expression of 2b, leads to a significant upregulation of m⁶A-marked genes.

Importantly, we noticed that most of these m⁶A-hypomethylated and upregulated genes are involved in plant defense responses (Supplementary Fig. 12a). For instance, the 36 genes with both reduced m⁶A and increased expression in CMV vs CMV-Δ2b comparison are believed to be directly regulated by CMV-2b (Fig. 7f, right panel, 7 g, and Supplementary Data 4). Among the 36 genes, the negative regulator genes of SA-mediated defense response, *CBP60a* and *NPR3*^{42,43}, were obviously demethylated around 3'UTR regions and upregulated following CMV infection but not CMV-Δ2b infection (Supplementary Fig. 12b). The specifically upregulated expression of these two genes by CMV infection were further confirmed by RT-qPCR (Supplementary Fig. 13a). We conducted m⁶A-IP-qPCR assays in Col-0 and *2b-Myc* transgenic plants to assess m⁶A modification levels. IgG-IP was included as a negative control. As shown in Fig. 7h, m⁶A levels in IgG-IP

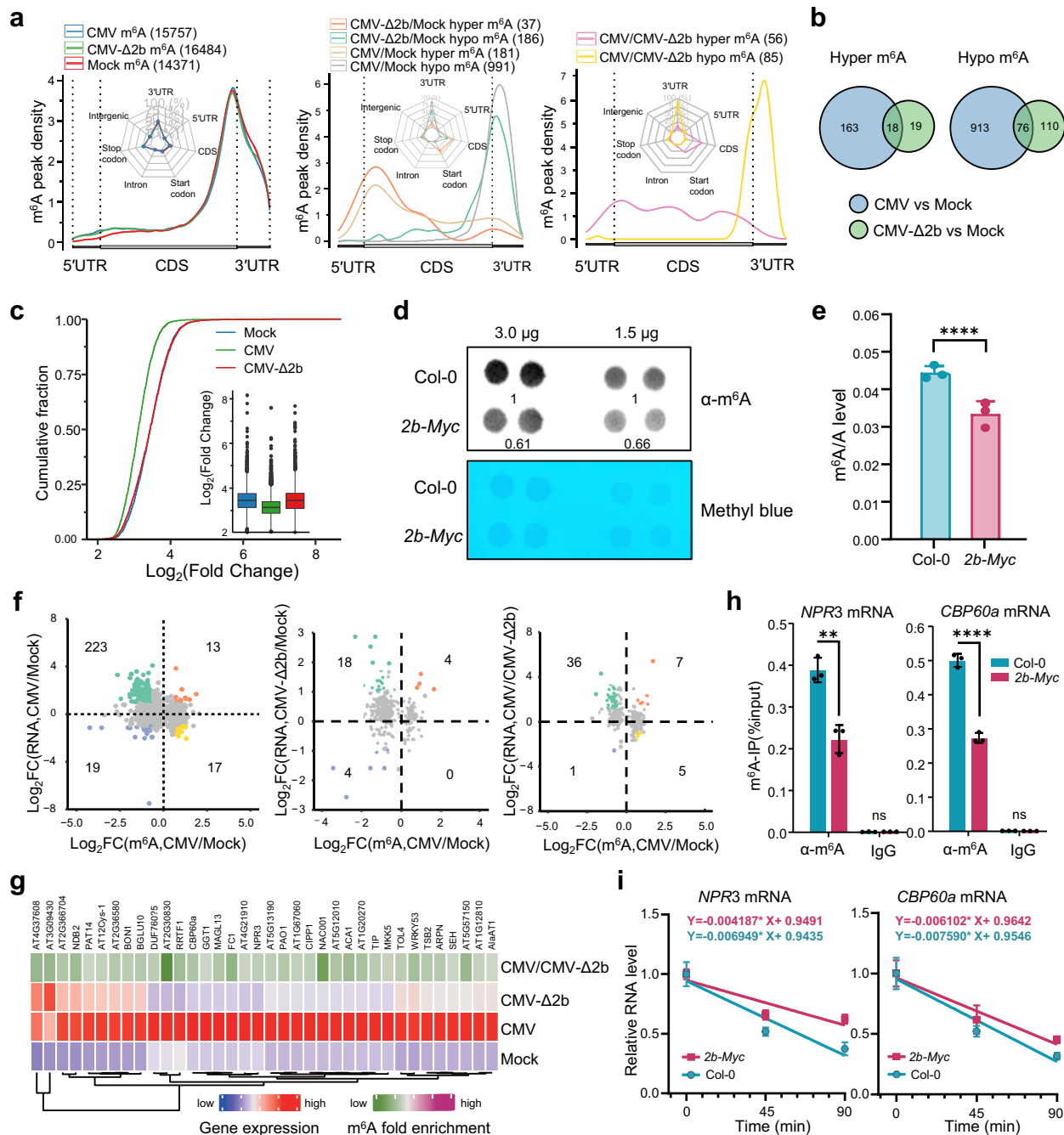


Fig. 7 | CMV-2b inhibits plant m^6A deposition. **a** Metagenomic and radar plots illustrate m^6A peak distributions identified in Col-0 across the indicated mRNA segments under mock, CMV-, and CMV-Δ2b-infected plants (left panel); hyper- and hypomethylated peaks relative to (middle); and differences between CMV and CMV-Δ2b infections (right), with m^6A peaks indicated in parentheses. **b** Venn diagram of genes with hyper-/hypo-methylated m^6A peaks in CMV- and CMV-Δ2b-infected plants versus mock. **c** Cumulative distribution (left) and boxplot (right) of m^6A enrichment ($\log_2[IP/Input]$) in mock-infected *A. thaliana*. The number of confident m^6A peaks (n) analyzed was 14,371 for mock, 15,757 for CMV, and 16,484 for CMV-Δ2b. For the boxplot, the center line denotes the median, box bounds represent the interquartile range (IQR), and whiskers extend to $1.5 \times IQR$. Data are the means $\pm$ SD ($n = 3$ biological independent replicates). Significance values were calculated with two-sided Student's t -test (mock vs. CMV, $p < 0.0001$; mock vs. CMV-Δ2b, $p = 0.05$; CMV vs. CMV-Δ2b, $p < 0.0001$). **d** Dot blot analysis revealing the enrichment of m^6A in Col-0 and transgenic *2b-Myc* plants. Two replicates were

loaded for each genotype. **e** LC-MS/MS quantification of the m^6A/A ratio in Col-0 and the *2b-Myc* transgenic plants. Data are the means $\pm$ SD ($n = 3$ biological independent replicates). Significance values were calculated with two-sided t -test ($*** p < 0.01$, $p = 0.0075$). **f** Conjoint analysis between m^6A modification and the expression profile of m^6A -containing genes in comparisons of CMV/mock (left), CMV-Δ2b/mock (middle), and CMV/CMV-Δ2b (right). **g** Heatmap showing the gene expression pattern and m^6A fold enrichment of the 36 m^6A hypomethylated genes in CMV-infected plants compared to CMV-Δ2b. *AT4G33380* serves as an internal control. Source data are provided as a Source Data file. **h** m^6A -IP qPCR showing reduced m^6A on *NPR3* and *CBP60a* mRNAs in *2b-Myc* plants compared to Col-0. Data are the means $\pm$ SD ($n = 3$ biological independent replicates). Significance values were calculated with two-sided t -test ($**** p < 0.0001$, $*** p < 0.01$, $p = 0.0031$). **i** mRNA stability of *NPR3* and *CBP60a* in Col-0 and *2b-Myc* plants; Data are the means $\pm$ SD ($n = 3$ biological independent replicates).

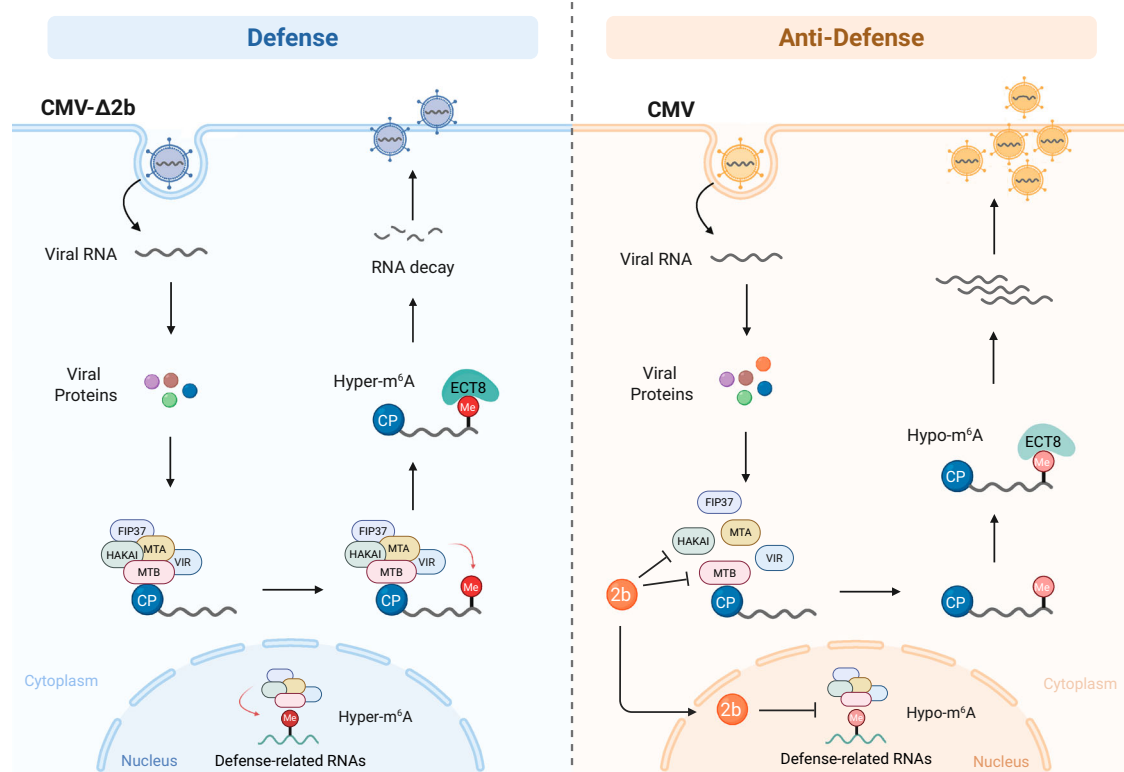


Fig. 8 | m⁶A-dependent antiviral defense and anti-defense mediated by viral 2b protein. During CMV-Δ2b infection, CP recruits the m⁶A catalytic complex to the cytoplasm via interaction with MTB, promoting the deposition of m⁶A modification on viral RNA. Hyper-methylated m⁶A on viral RNA is recognized by ECT8, promoting the degradation of viral RNA, thereby inhibiting the amplification of the virus and playing a defensive role. When CMV infects, the 2b protein disrupts the

functionality of m⁶A complex by binding to MTB and HAKAI, inhibiting m⁶A deposition on viral RNA and nuclear defense-related RNA (such as *CBP60a* and *NPR3*), enhancing the accumulation of viral RNA and immune negative regulators, thereby inhibiting plant antiviral immunity. Figure created in BioRender. Duan, C. (2025) <https://BioRender.com/31hdjjs>.

samples were negligible and did not differ between Col-0 and *2b-Myc*, indicating the specificity and reliability of the m⁶A-IP assay. Importantly, m⁶A levels in m⁶A-IP samples were significantly reduced in *2b-Myc* plants, consistent with our MeRIP-seq results, confirming that CMV 2b inhibits m⁶A deposition on key immune-related transcripts, such as *CBP60a* and *NPR3*.

The reduction of m⁶A and increase in gene expression prompt us to hypothesize that m⁶A might negatively affect the stability of these RNAs. To test the functional relevance of this regulation, we performed mRNA decay assays and found that *CBP60a* and *NPR3* transcripts were significantly more stable in *2b-Myc* transgenic plants than in wild-type Col-0 (Fig. 7i). These findings suggest that CMV 2b promotes the stability of *CBP60a* and *NPR3* mRNAs by interfering with m⁶A-mediated RNA decay. Supporting this notion, both *cbp60a* and *npr3* mutants displayed increased resistance to CMV infection (Supplementary Fig. 13b–d).

Collectively, we propose a working model summarizing the interplay between m⁶A modification and CMV infection (Fig. 8). During CMV-Δ2b infection, the viral CP protein recruits m⁶A writers to the cytoplasm via interaction with MTB, facilitating m⁶A deposition on viral RNAs. These m⁶A-hypermethylated viral RNAs are recognized by ECT8, promoting viral RNA decay and suppression of viral replication. Simultaneously, in the nucleus, m⁶A modification can also be deposited at defense-related RNAs to regulate their stability. However, in wild-type CMV infection, the 2b protein disrupts the functionality of m⁶A writer complex by competitively binding to MTB and HAKAI. This results in inhibition of m⁶A deposition on both viral RNAs and defense-related nuclear mRNAs, leading to enhanced stability of viral RNA and altered expression of defense-related mRNAs.

Discussion

Plants have evolved sophisticated defense mechanisms to combat threats posed by pathogenic microorganisms. Beyond innate immune mechanisms, RNA-based immunity provides an additional, rapid, and broad-spectrum defense system. In response, pathogens have developed diverse counterstrategies, including antagonistic mechanisms targeting RNAi pathways. A well-studied example is the mutual antagonism between the plant antiviral RNAi pathway and virus-encoded VSRs. This RNA-based antagonism also extends to plant interactions with non-virus pathogens. A recent study has shown that plants can deliver endogenous sRNAs into the pathogenic fungus *Verticillium dahliae*, inducing trans-kingdom RNAi and silencing virulence genes⁴⁴. Conversely, *V. dahliae* counters this defense by secreting effector proteins into plant cells, thereby disrupting the nuclear export of plant sRNAs and impairing trans-kingdom RNAi immunity⁴⁴. As the most prevalent internal modification in eukaryotic mRNAs, m⁶A has been shown to regulate numerous biological processes, including antiviral responses in plants. However, whether a mutual antagonism exists between plant m⁶A machinery and plant viruses remains unclear. In this study, we uncover a previously unrecognized novel m⁶A-mediated antagonistic mechanism between plants and the highly destructive RNA virus CMV. We demonstrate that plant m⁶A methyltransferases are recruited to viral RNA through direct interaction with viral CP protein in the cytoplasm, facilitating the deposition of m⁶A modification. These modifications are then recognized by the reader protein ECT8, which promotes the degradation of hypermethylated viral RNAs, thereby enhancing plant resistance. CMV counters this mechanism through its VSR protein 2b, which disrupts the functionality of m⁶A writer complex by interfering with MTB-HAKAI interaction,

leading to reduced m⁶A deposition and increased stability of viral RNA, ultimately facilitating viral infection.

While m⁶A has long been implicated in the regulation of animal viral pathogenicity⁴⁵, its role in plant-virus interactions has only recently begun to emerge. In 2017, Martinez-Perez et al. first reported the presence of m⁶A modifications on the genomic RNAs of AMV and CMV⁹. Subsequent studies have identified m⁶A modifications in several other plant viruses^{23–25,46,47}, with most indicating that m⁶A modification promotes viral RNA decay, consistent with our findings. Notably, an exception was observed in wheat, where the *Triticum aestivum* methyltransferase TaMTB promotes m⁶A deposition on wheat yellow mosaic virus (WYMV), enhancing its stability and virulence²⁵.

It is well established that the canonical m⁶A methyltransferase complex primarily functions within the nucleus. Our data also demonstrated that MTB, FIP37, and HAKAI are predominantly localized to the nucleus in mock conditions, although MTA and HAKAI displayed substantial cytoplasmic distribution (Fig. 3a). Given that CMV genomic RNA replicates in the cytoplasm, a key question is how viral RNAs are targeted by these nucleus-localized methyltransferases. Here, we reveal that during CMV infection, the core methyltransferase MTB interacts with CMV-CP protein in the cytoplasm, recruiting other methyltransferases (MTA, HAKAI, and FIP37) from the nucleus to the cytoplasm, thereby facilitating m⁶A deposition on viral RNAs.

CMV-2b is one of the earliest identified viral VSRs and employs multiple mechanisms to enhance pathogenicity, chiefly through suppression of antiviral RNAi. It binds dsRNA to block the production, loading, and spreading of vsRNAs¹³. Additionally, it interacts with Argonaute 1 (AGO1) to disrupt its slicing activity⁴⁸. The powerful suppressive effect of VSRs like 2b often obscures genotypic differences in antiviral responses, complicating the identification of novel antiviral factors. Consequently, VSR-deficient viruses such as CMV-Δ2b are often employed for genetic screening of antiviral factors. For example, the aminophospholipid-transporting ATPase 1 (ALA1) and ALA2 were identified as key components of antiviral RNAi through a forward genetic screen of CMV-Δ2b^{49,50}. In addition, mutants deficient in RNAi (e.g., *dcl2/4*, *rdr1/6*, *sgs3*) exhibited significantly increased susceptibility to CMV-Δ2b compared to wild-type CMV^{29,49,51}. In accordance, we found that both m⁶A methyltransferases and demethylase mutants displayed differential susceptibility to CMV-Δ2b but not to CMV, suggesting that CMV-2b can effectively suppress m⁶A-mediated antiviral defense. This may explain why no significant differences in CMV susceptibility was observed in *alkb9b* mutants (m⁶A demethylase) in an earlier study reported by Martinez-Perez et al.⁹. In our study, we found that CMV-2b directly interacts with HAKAI and MTB, disrupting m⁶A deposition on viral RNA and thereby impairing this layer of antiviral defense. A parallel mechanism was observed in PepMV, whose RdRP exploits autophagy pathway to degrade *Solanum lycopersicum* HAKAI, indirectly inhibiting m⁶A-mediated antiviral defense²⁴.

Notably, the production of vsRNAs during the infection of both CMV and CMV-Δ2b was not significantly affected in m⁶A-deficient mutants (Fig. 1c), suggesting that increased viral susceptibility is not a result of compromised antiviral RNAi pathway. Previous research has shown that MTA-mediated m⁶A deposition on primary miRNA transcripts (pri-miRNA) alters their secondary structure and impairs pri-miRNA processing, ultimately leading to reduced accumulation of mature miRNAs⁵². Moreover, recent studies also indicate that the m⁶A methyltransferases facilitate the recruitment of miRNA processing machinery via MTB-SERRATE interactions³³. Given that vsRNAs and miRNAs are processed from different dsRNA substrates by different processors in separate cellular compartments, it is plausible that vsRNA biogenesis remains unaffected in m⁶A-deficient mutants.

In Arabidopsis, there are 13 predicted m⁶A reader proteins. Among them, ECT2/3/4 regulate leaf morphogenesis and abscisic acid responses. Recent studies have identified ECT8 as a key player in abscisic acid (ABA) perception and abiotic stress response^{36,37}. In

response to ABA, ECT8 forms phase-separated condensates in stress granules to sequester m⁶A-modified ABA receptor gene *PYRABACTIN RESISTANCE1-LIKE7* (*PYL7*) mRNA, inhibiting its translation and attenuating ABA perception³⁷. More recently, Cai et al. demonstrated that ECT8 acts as an abiotic stress sensor³⁶. ECT8 is significantly induced by various environmental stresses, and the increased abundance results in enhanced binding capability to m⁶A-modified mRNA, accelerating their degradation through direct interaction with the decapping protein DCP5 within processing bodies. Thus, ECT8 is vital for the regulation of transcriptome homeostasis during abiotic stress response. In our study, CMV infection significantly induced the expression of *ECT7*, *ECT8*, and *ECT10*, and mutants of these genes showed heightened susceptibility to CMV-Δ2b (Fig. 5b), implicating them in plant defense against the virus. In line with this notion, ECT8 displayed specific binding capability to m⁶A-modified viral RNAs in vivo and in vitro, confirming its role as a reader of viral m⁶A. Given that ECT8 facilitates the decay of m⁶A-modified mRNA, this conclusion was further supported by the observation that the stability of m⁶A-modified viral RNA was significantly reduced (Fig. 2c). Intriguingly, ECT8 condensates were not visibly formed during CMV infection (Supplementary Fig. 14), possibly due to insufficient viral titers or a mechanistic divergence in how ECT8 interacts with viral versus endogenous m⁶A-modified mRNAs. The potential involvement of ECT7 and ECT10 as additional viral m⁶A readers warrants further exploration.

Methods

Plant materials and growth conditions

All Arabidopsis materials used in this study were derived from the Columbia-0 (Col-0) accession. T-DNA insertion mutants, including *hakai-1* (GK_217A12), *ect1* (SALK_073479), *ect2* (SAIL_1174_G01), *ect3* (SALK_134503), *ect4* (SALK_112012), *ect5* (SALK_111736), *ect6* (SALK_083381), *ect7* (SALK_105881), *ect8* (SALK_206710), *ect10* (SAIL_357_C10), *ect11* (SALK_084901), *ect12* (SALK_071944), *alkb9b* (SALK_015591), *npr3* (SALK_043055), and *cbp60a* (SALK_124410) were obtained from the Arabidopsis Biological Resource Center. CRISPR/Cas9-mediated mutagenesis was employed to generate *hakai-c* and *alkb10b-c* mutants in the Col-0 background, as illustrated in Supplementary Fig. 1. The *ABI3:MTA* mutant⁵⁴ and the *dcl2/4*⁴⁹ mutant were characterized in previous studies. Transgenic plants *HAKAIcom* were generated by transforming the *hakai-c* mutant with a *HAKAIpro:gHAKAI-Flag* construct. The *Flag-MTB* line was produced by transforming Col-0 with a *ACTIN2pro:Flag-MTB-CDS* construct, and the *HAKAI-Myc* line was produced by transforming Col-0 with a *35Spro:HAKAI-CDS-Myc* construct. The *2b-Myc* line was produced by transforming Col-0 with a *35Spro:2b-Myc* construct. The *ECT8pro:YFP-gECT8* transgenic line was generated by transforming *ECT8pro:YFP-gECT8* construct into *ect8* mutant. Both *MTBpro:MTB-GFP* and *FIP37pro:FIP37-GFP* transgenic plants are gifts from Dr. Ruiqiang Ye's Laboratory in the Center for Excellence in Molecular Plant Science, Chinese Academy of Sciences. Homozygous T3 transgenic plants were selected and propagated for experimental analyses. Arabidopsis plants were cultivated either on 0.5× Murashige and Skoog (MS) agar plates or in soil, under long-day conditions (16 h light/8 h dark) at 22 °C. *N. benthamiana* plants were grown in soil under the same photoperiod at 25 °C. Virus-infected plants were maintained under short-day conditions (8 h light/16 h dark) at 22 °C. All primers used in this study are listed in Supplementary Data 5.

Viral particle purification and virus inoculation assays

The viral strains used in this study belong to CMV subgroup IB, specifically the China ShanDong strain (SD-CMV). The wild-type CMV and CMV-Δ2b have been thoroughly characterized in a previous study²⁶. Viral particle purification and inoculation assays were carried out according to the established method⁵⁵. Specifically, leaf tissues from systemically infected *A. thaliana* or tobacco plants were collected, carefully

excluding stems and main veins. The tissues were flash-frozen in liquid nitrogen and ground into a fine powder. For each gram of ground tissue, 1 mL of ice-cold Buffer A (0.5 M sodium citrate (pH 6.5–7.0), 5 mM EDTA, 0.5% (v/v) thioglycolic acid (freshly added)) and 1 mL of chloroform were added. The mixture was gently stirred at 4 °C for 1–2 h and centrifuged at $15,000 \times g$ for 10 min at 4 °C to pellet cellular debris. The supernatant was filtered through Miracloth (Millipore, 475855) and evenly aliquoted into ultracentrifuge tubes, leaving at least 5 mL of headspace in each. Using extra-long pipette tips, 5 mL of ice-cold Buffer A containing 10% (w/v) sucrose was carefully underlaid beneath each sample by slow dispensing to the tube bottom. Samples were ultracentrifuged at $102,000 \times g$ for 2 h at 4 °C. Following centrifugation, the supernatant was gently decanted and discarded. The pellet was resuspended in 3–5 mL of Buffer B (5 mM sodium borate (pH 9.0), 0.5 mM EDTA) and incubated overnight at 4 °C to facilitate particle recovery. This suspension was clarified by centrifugation at $10,000 \times g$ for 15 min at 4 °C. The final supernatant, containing purified viral particles, was collected, and viral concentration was determined by measuring absorbance at 260 nm (A_{260}) using a Nanodrop spectrophotometer. An A_{260} reading of 5.0 was equivalent to 1 mg/mL of viral particles. Three-week-old *A. thaliana* plants grown under short-day conditions were inoculated on the largest true leaves with either wild-type CMV or the CMV-Δ2b mutant. Viral stocks were diluted in Buffer B to a final concentration of 20 ng/μL. Approximately 10 μL of inoculum was applied to each of three expanded leaves per plant. Silicon carbide (carborundum) was dusted onto the leaves, followed by gentle mechanical rubbing with a gloved fingertip to ensure successful inoculation. For tobacco infection assay, *Agrobacterium* containing CMV infectious clones were infiltrated into *N. benthamiana* leaves²⁶.

RNA extraction and gel blotting analysis

Total RNA was extracted utilizing TRIzol reagent (Invitrogen, 15596018), while low molecular weight RNA (LMW) was isolated via a hot phenol method⁴³. Specifically, LMW RNA was extracted from virus-infected plant tissue as follows: Approximately 2 g of fresh leaf tissue was ground to a fine powder in liquid nitrogen using a pre-chilled mortar and pestle. Four milliliters of phenol buffer, preheated to 75 °C, was added to the powder, followed by immediate and vigorous vortexing. One milliliter of chloroform was then added, and the mixture was vortexed for 20 seconds and incubated at room temperature for 10 min. Phase separation was performed by centrifugation at $7500 \times g$ for 20 min at 4 °C. The aqueous phase was transferred to a new tube and mixed with an equal volume of 4 M lithium chloride to selectively precipitate high molecular weight RNA. The mixture was incubated overnight at –20 °C and centrifuged at $7500 \times g$ for 20 min at 4 °C. The resulting supernatant was transferred to a fresh tube, and RNA was precipitated by adding two volumes of absolute ethanol and one-tenth volume of 3 M sodium acetate (pH 5.2). After thorough mixing, the solution was incubated at –20 °C for 3 h. The RNA was pelleted by centrifugation at $7500 \times g$ for 20 min at 4 °C, washed twice with 75% ethanol (centrifuged at $7500 \times g$ for 5 min at 4 °C each time), and air-dried at room temperature. The final RNA pellet was resuspended in an appropriate volume of DEPC-treated water.

Viral RNA and small RNA blotting were conducted following the protocol in previous studies^{55,56}. For viral RNA detection, 10 μg of total RNA was separated on a 1.2% (w/v) agarose denaturing gel containing 0.1% (v/v) formaldehyde. Following gel electrophoresis, the RNA was transferred onto Hybond-N+ membranes (GE Healthcare, RPN303B) via UV cross-linking. For vsiRNA detection, 20 μg of low molecular weight RNA was resolved on a 15% polyacrylamide gel and transferred to Hybond-NX membranes (GE Healthcare, RPN303T) using 1-ethyl-3-(3-dimethylaminopropyl) (Thermo Scientific, 22981)-mediated cross-linking⁵⁷. Hybridization was performed using biotin-labeled DNA probes, and signal detection was performed with the

Chemiluminescent Nucleic Acid Detection Module Kit (Invitrogen, 11789020). Probe sequences are provided in Supplementary Data 5.

Protein expression, purification, and EMSA

The coding sequences of *ECT8* and 2b were cloned into the pGEX-4T-1 vector and expressed in *E. coli* BL21 Gold cells. Protein expression was induced with 0.4 mM isopropyl β-d-1-thiogalactopyranoside (IPTG) at 16 °C overnight once the culture reached $OD_{600} = 0.6$ –0.8. GST, GST-fused ECT8 and 2b proteins were purified using Glutathione Sepharose (GE Healthcare, 17-0756). The EMSA assays were performed using a established protocol with minor modification^{58,59}. Briefly, 5 pmol of Cy3-labeled RNA probes were incubated with varying concentrations of GST-ECT8 or GST in binding buffer (10 mM HEPES (pH 8.0), 50 mM KCl, 1 mM EDTA, 0.05% (v/v), Triton-X-100, 5% (v/v) glycerol, 1 mM DTT, 10 mg/mL salmon DNA (Sigma-Aldrich, D7656), 1× protease inhibitor cocktail (APEX BIO, K1011) and 40 U/mL RNasin (APEX BIO, K1046)) for 60 min on ice. The samples were then resolved on a 4% native polyacrylamide gel, and fluorescence signals were visualized using an Amersham Typhoon 5 machine (Cytiva).

Subcellular localization and BiFC analysis

For *Arabidopsis* transgenic lines stably expressing MTB, FIP37, and ECT8 fused to GFP, fluorescence signals in systemically infected leaves were monitored at various time points following viral or mock infection. For transient expression in tobacco, the coding sequences of CMV-2b, CP, *Arabidopsis* MTB, HAKAI, and FIP37 were cloned into the pGDR and pGDG vectors and transformed into the *Agrobacterium* strain GV3101. Cultures carrying recombinant constructs were suspended in infiltration buffer (10 mM MgCl₂, 10 mM MES (pH 5.6), and 0.2 mM Acetosyringone) and infiltrated into the 4-week-old *N. benthamiana* leaves. After 48 h, fluorescence signals were detected using a Leica Stellaris SR5 laser scanning confocal microscope.

For BiFC assay, split YFP fusion constructs were generated for each gene of interest. *N. benthamiana* leaves were pre-infiltrated with CMV infectious clone. At 4 dpi, *Agrobacterium* containing split YFP constructs were infiltrated into either virus-free or CMV-infected *N. benthamiana* leaves. YFP signal was observed and quantified 48 h post-infiltration.

Protein interaction analysis

For Y2H assay, the coding sequences of CMV proteins and the *Arabidopsis* proteins MTA, MTB, HAKAI, and FIP37 were cloned into the pGADT7 and pGBKT7 vectors to generate AD and BD constructs, respectively. These constructs were transformed into the yeast strain AH109. Transformants were first selected on SD medium lacking leucine and tryptophan (–L/T), and then screened on SD medium lacking leucine, tryptophan, histidine, and adenine (–L/T/H/A) to assess protein-protein interactions. Plates were incubated at 30 °C for 3 days.

For the Y3H assay, the coding sequence of MTB was inserted into the MCS I site of the pBridge vector, while the coding sequence of CMV-2b was inserted into the MCS II site under the control of the methionine-repressible MET promoter, serving as a bridge protein. The pGADT7-HAKAI and pBridge-MTB-2b constructs were co-transformed into the AH109 strain. Transformants were plated on SD medium lacking leucine, tryptophan, and methionine (–L/T/M); SD medium lacking leucine, tryptophan, histidine, and methionine (–L/T/H/M); and SD medium lacking leucine, tryptophan, and histidine (–L/T/H) to evaluate ternary protein interactions. Plates were incubated at 30 °C for 3 days.

For the Co-IP assay, the coding sequences of *MTA*, *MTB*, *HAKAI*, *FIP37*, and CMV-2b were individually cloned into binary vectors harboring a CaMV 35S or *ACTIN2* promoter and either Myc or Flag tag. *Agrobacterium* strains containing these constructs were infiltrated into 4-week-old *N. benthamiana* leaves. At 48 h post-infiltration, the leaves were harvested and total proteins were extracted using lysis

buffer (25 mM Tris-HCl (pH 7.5), 150 mM NaCl, 1 mM EDTA (pH 8.0), 10% (v/v) glycerol, 0.15% (v/v) NP-40, 1 mM DTT, 0.1 mM PMSF, and 1× protease inhibitor cocktail (APEX BIO, K1011)). Protein extracts were incubated with Dynabeads Protein G (Invitrogen, 10004D) pre-coupled with anti-Myc (Millipore, 05-724) or anti-Flag M2 (Sigma-Aldrich, F1804) antibodies at 4 °C for 3 h. Following four washes with the lysis buffer, the bound proteins were eluted and analyzed via immunoblotting to detect specific interactions. For the co-IP assay performed in Col-0 and *HAKAI-Myc*, and *Flag-MTB* plants, plants were harvested 14 days after viral infection, and the same procedure was employed. Endogenous anti-CP antibody was kept in our laboratory. Endogenous anti-MTB antibody has been published⁶⁰.

RNA stability analysis

Twelve-day-old seedlings of Col-0 and *2b-Myc* plants, as well as Col-0, *hakai-c*, and *alkb10b-c* leaves systemically infected with CMV for 7 days, were subjected to vacuum infiltration with 0.6 mM cordycepin in infiltration buffer (1 mM PIPES (pH 6.25), 1 mM sodium citrate, 1 mM KCl, and 15 mM sucrose) for 5 min, followed by incubation at room temperature for 15 min. Leaf samples were then harvested at 0, 60, and 120 min post-treatment and immediately flash-frozen in liquid nitrogen. Total RNA was extracted using TRIzol reagent (Invitrogen, 15596018), and cDNA synthesis was performed with the HiScript III 1st Strand cDNA Synthesis Kit (Vazyme, R312-02). Quantitative real-time PCR (RT-qPCR) was performed using AceQ qPCR SYBR Green Master Mix (Vazyme, Q111-03) on a CFX96 Touch Deep Well Real-Time PCR Detection System (Bio-Rad).

m⁶A-IP-qPCR Analysis

Three grams of *A. thaliana* seedlings (either 12 days old or harvested two weeks post-CMV infection) were collected for RNA extraction using TRIzol reagent (Invitrogen, 15596018). Total RNA was then subjected to two rounds of poly(A) + RNA enrichment using Dynabeads Oligo (dT). The enriched RNA was fragmented into ~200 nt fragments using the NEB Next Magnesium RNA Fragmentation Module (NEB, E6150S). A small portion of the fragmented RNA was reserved as input, while the remaining RNA was incubated overnight at 4 °C with anti-m⁶A antibody (Synaptic Systems, 202111) or control IgG (Abclonal, AC005) in IP buffer (10 mM Tris-HCl (pH 7.5), 150 mM NaCl, 0.1% (v/v) NP-40, 40 U/mL RNasin (APEX BIO, K1046)). Following incubation, Dynabeads Protein G (Invitrogen, 10004D) were added for 2 h at 4 °C to perform immunoprecipitation. Beads were then washed four times with IP buffer. Both immunoprecipitated RNA (IP RNA) and input RNA were purified and subsequently analyzed by RT-qPCR.

RIP-qPCR analysis

RIP-qPCR was performed using systemic leaves from *ECT8pro:YFP-gECT8* transgenic *Arabidopsis* plants, harvested at 14 dpi. The procedure was based on a previously published protocol³⁵, with minor modifications. Specifically, formaldehyde crosslinking was carried out by adding formaldehyde to 2 g of freshly harvested leaf tissue to a final concentration of 1%, followed by vacuum infiltration for 20 min at 4 °C. The crosslinking reaction was quenched by adding glycine to a final concentration of 125 mM via vacuum infiltration for 15 min at 4 °C. The tissue was then washed four times with DEPC-treated water and blotted dry with filter paper. The crosslinked tissue was ground in liquid nitrogen using a mortar and pestle, then resuspended in 4 mL of RIP buffer (50 mM Tris-HCl (pH 8.0), 200 mM NaCl, 5 mM MgCl₂, 0.2% (v/v) NP-40, 10% (v/v) glycerol, 1 mM PMSF, 0.5 mM DTT, 1× protease inhibitor cocktail (APEX BIO, K1011), and 40 U/mL RNasin (APEX BIO, K1046)). The lysate was incubated on ice for 15 min, followed by centrifugation at 7500 × g for 20 min at 4 °C. The supernatant was filtered through four layers of Miracloth (Millipore, 475855) to remove plant debris. Protein-RNA mixture was fragmented by sonication using a Bioruptor Plus (Diagenode) set to

low power, with 30-second ON/30-second OFF cycles in a 4 °C water bath. Sonication was performed for a total of 12–15 min to achieve fragments of ~200 nt. The sonicated lysate was centrifuged at 16,000 × g for 10 min at 4 °C, and the resulting supernatant was transferred to fresh tubes. A small aliquot was saved as input, and the remainder was incubated overnight at 4 °C with an anti-GFP antibody (Abmart, M20004). Dynabeads Protein G (Invitrogen, 10004D) were added to the antibody-bound lysates and incubated for 2 h at 4 °C to perform immunoprecipitation. Beads were washed four times with RIP wash buffer (20 mM Tris-HCl (pH 8.0), 150 mM NaCl, 2 mM EDTA, 0.1% (v/v) Triton X-100, 0.1% (v/v) SDS, 1 mM PMSF, 1× protease inhibitor cocktail (APEX BIO, K1011), 5 U/mL DNase I (Takara, 2270 A), and 40 U/mL RNasin (APEX BIO, K1046)). Protein-RNA complexes were digested with Proteinase K (Thermo Scientific, AM2546), and RNA was purified from both input and immunoprecipitated fractions. The recovered RNA was analyzed by RT-qPCR.

Dot blot analysis of m⁶A enrichment

To investigate m⁶A enrichment in plants and viral genomes, total RNA was extracted from plant tissues, and viral genomic RNA was isolated from purified viral particles. Equal amounts of RNA were UV-crosslinked onto Hybond-N+ membranes (GE Healthcare, RPN303B). The membranes were subsequently probed with an m⁶A-specific antibody (Synaptic Systems, 202111) for detection.

MeRIP-seq assay and data analysis

For RNA-seq and MeRIP-seq assays, 24-day-old Col-0 plants were mechanically inoculated by rubbing with freshly prepared sap from CMV or CMV-Δ2b-infected *Arabidopsis* leaves. Mock-treated, CMV-, and CMV-Δ2b-infected plants were collected two weeks after inoculation for downstream analysis. Total RNA was extracted using TRIzol reagent (Invitrogen, 15596018). For MeRIP-Seq, 50 μg of total RNA from each sample was subjected to two rounds of poly(A) RNA enrichment using Dynabeads Oligo(dT). The purified poly(A) RNA was fragmented into 200–500 nt fragments using the Magnesium RNA Fragmentation Module (NEB, E6150S). The fragmented RNA was immunoprecipitated using an anti-m⁶A antibody (Synaptic Systems, 202111), and the enriched RNA was used for library construct following the Illumina protocol. Briefly, the m⁶A-enriched RNA fragments underwent end repair with a mixture of enzymes (*E. coli* DNA polymerase I, RNase H, and dUTP), followed by addition of an A-tail to the blunt ends for ligation to indexed adapters with T-overhangs. The indexed adapters were then ligated to the fragment. Library size selection was performed with AMPureXP beads (BECKMAN COULTER, A63882). After heat-labile UDG enzyme treatment to remove U-labels from the second-stranded DNAs, the ligated products were PCR amplified. The resulting cDNA libraries, with an average insert size of 300 ± 50 bp, were sequenced on an Illumina Novaseq™ 6000 platform using 150 bp paired-end mode (PE150).

For MeRIP-seq data analysis, adapter sequences and low-quality bases reads were trimmed using Cutadapt v1.18¹. Reads with ambiguous nucleotides or shorter than 15 nucleotides were filtered out by Trimmomatic v0.30⁶¹. Clean reads were mapped to the *Arabidopsis thaliana* TAIR10 reference genome using Burrows-Wheeler Aligner⁶² (BWA version 0.30). Only uniquely mapped reads with a mapping quality (MAPQ) ≥ 13 were retained for subsequent analysis⁶³. High-confidence m⁶A peaks were identified using R package exomePeak2⁶⁴. Differential methylated peaks were identified by MeTDiff⁶⁵, using a fold change threshold > 1.5 and a false discovery rate (FDR) < 0.05. In addition, MACS software (version 2.0.10)⁶⁶ was used to call m⁶A peaks from the IP samples using corresponding input samples as controls. R packages (ChIPPeakAnno) was employed to annotate confident peaks to the *Arabidopsis* annotation file. GO enrichment analysis of m⁶A-modified genes was performed using GOATOOLS⁶⁷.

For RNA-seq analysis, input reads from the MeRIP-seq dataset were processed similarly. Briefly, clean reads were aligned using STAR⁶⁸, and gene expression levels were quantified as fragments per kilobase of transcript per million mapped reads (FPKM) using RSEM⁶⁹. Differentially expressed genes were identified by R package (edgeR)⁷⁰, with a fold change ≥ 1.5 and a p value ≤ 0.05 .

LC-MS/MS analysis of m⁶A enrichment

RNA digestion was performed by incubating 200 ng RNA with 1 unit of Nuclease P1 (NEB, M0660S) in 20 μ L of buffer (25 mM NaCl, 25 mM ZnCl₂) at 37 °C and 60 $\times g$ for 2 h. Subsequently, 2 μ L of 2 M NH₄HCO₃ and 1 μ L of Shrimp Alkaline Phosphatase (NEB, M0371V) were added, and incubation continued at 37 °C and 60 $\times g$ for an additional 2 h. The reaction was neutralized by adding 1 μ L of 1.2 μ M HCl. The sample was then centrifuged at 16,000 $\times g$ for 30 min at 4 °C, and the aqueous phase was collected.

Quantitative LC-MS analyses of m⁶A and adenosine(A) were performed by a QTRAP mass spectrometer (6500, AB Sciex) equipped with UHPLC (Acquity, Waters) and an XSelect HSS T3 column (100 mm $\times$ 3.0 mm, 2.5 μ m, Waters). The injection volume was 1 μ L. The mobile phase A was 0.1% formic acid in water, and B was methanol. The column was maintained at 40 °C with a flow rate of 0.4 mL/min, and the gradient of B was as follows: 0 min, 2%; 0.5 min, 2%; 8 min, 20%; 9 min, 99%; 13 min, 99%; 13.1 min, 2%; 17 min, 2%. All analytes were detected using Multiple reaction monitoring (MRM) mode. The optimized ESI operating parameters for positive mode were: ion spray voltage, 4.5 kV; ion spray temperature, 450 °C; curtain gas, 35 psi; ion source gas 1, 50 psi; ion source gas 2, 50 psi. Nucleosides were quantified based on the transition of the parent ribonucleoside to the deglycosylated base ion: m/z 268.1–136.1 for A, m/z 282.1–150.1 for m⁶A. Absolute quantities of each ribonucleoside were determined using an external calibration curve prepared with A (Selleck, S1647), m⁶A (Selleck, S3190) standards. Instrument control and data acquisition were performed using Analyst 1.6.3 software (AB SCIEX), and data processing was performed using MultiQuant 3.0.2 software (AB SCIEX).

Nanopore DRS and bioinformatics analysis

Total RNA was extracted from *A. thaliana* Col-0 plants 14 days post-inoculation with either CMV or CMV- Δ 2b. RNA quality was rigorously assessed through visual inspection, agarose gel electrophoresis, and quantification using both Nanodrop and Qubit fluorometry. To enrich for viral transcripts, polyadenylated mRNA was purified using Oligo(dT) magnetic beads. Reverse transcription was initiated by ligating a reverse transcription adapter and synthesizing a hybrid strand, followed by magnetic bead-based purification. Sequencing adapters were then ligated to the purified cDNA using the Oxford Nanopore Technologies (ONT) SQK-RNA004 kit. The resulting cDNA libraries were quantified with Qubit and loaded onto an ONT PromethION 24 flow cell for real-time, single-molecule direct RNA sequencing, allowing for the detection of native m⁶A modifications in viral transcripts.

For downstream analysis, the CMV reference genome (DNA) was converted to an RNA sequence using seqkit seq-dna2rna⁷¹. Direct RNA sequencing reads were aligned to this custom viral transcriptome reference using Minimap⁷², generating a BAM file that was subsequently sorted and indexed using Samtools⁷³. m⁶A modification profiles at the transcript level were then extracted using Modkit pileup (<https://github.com/nanoporetech/modkit>).

Statistical analysis

All statistical comparisons of group means, as illustrated in the box plots, were performed using the two-sided Student's t-test. All tests were conducted on data from three independent biological replicates ($n=3$). Specific P -values for pairwise comparisons are indicated directly within the figures or figure legends. A P -value < 0.05 was adopted as the threshold for statistical significance.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The MeRIP-Seq and RNA-Seq data generated in this study have been deposited in the NCBI Gene Expression Omnibus (GEO) under accession code: [GSE269979](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE269979). The DRS data in this study have been deposited in NCBI GEO under accession code: [GSE293986](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE293986). Other data that support the findings of this study are available in the published article and its Supplementary Information file. Source data are provided with this paper.

References

- Cohn, W. E. Pseudouridine, a carbon-carbon linked ribonucleoside in ribonucleic acids: isolation, structure, and chemical characteristics. *J. Biol. Chem.* **235**, 1488–1498 (1960).
- Sharma, B., Prall, W., Bhatia, G. & Gregory, B. D. The diversity and functions of plant RNA modifications: what we know and where we go from here. *Annu. Rev. Plant Biol.* **74**, 53–85 (2023).
- Desrosiers, R., Friderici, K. & Rottman, F. Identification of methylated nucleosides in messenger RNA from Novikoff hepatoma cells. *Proc. Natl Acad. Sci. USA* **71**, 3971–3975 (1974).
- Zhong, S. et al. MTA is an Arabidopsis messenger RNA adenosine methylase and interacts with a homolog of a sex-specific splicing factor. *Plant Cell* **20**, 1278–1288 (2008).
- Zhang, M. et al. Two zinc finger proteins with functions in m(6)A writing interact with HAKAI. *Nat. Commun.* **13**, 1127 (2022).
- Ruzicka, K. et al. Identification of factors required for m(6) A mRNA methylation in Arabidopsis reveals a role for the conserved E3 ubiquitin ligase HAKAI. *N. phytologist* **215**, 157–172 (2017).
- Scutenaire, J. et al. The YTH domain protein ECT2 is an m(6)A reader required for normal trichome branching in arabidopsis. *Plant Cell* **30**, 986–1005 (2018).
- Duan, H. C. et al. ALKBH10B is an RNA N(6)-methyladenosine demethylase affecting arabidopsis floral transition. *Plant Cell* **29**, 2995–3011 (2017).
- Martinez-Perez, M. et al. Arabidopsis m(6)A demethylase activity modulates viral infection of a plant virus and the m(6)A abundance in its genomic RNAs. *Proc. Natl Acad. Sci. USA* **114**, 10755–10760 (2017).
- Tang, J. et al. The RNA N(6)-methyladenosine demethylase ALKBH9B modulates ABA responses in Arabidopsis. *J. Integr. Plant Biol.* **64**, 2361–2373 (2022).
- Lopez-Gomollon, S. & Baulcombe, D. C. Roles of RNA silencing in viral and non-viral plant immunity and in the crosstalk between disease resistance systems. *Nat. Rev. Mol. Cell Biol.* **23**, 645–662 (2022).
- Cui, C. et al. A brassica miRNA regulates plant growth and immunity through distinct modes of action. *Mol. Plant* **13**, 231–245 (2020).
- Duan, C. G. et al. Suppression of Arabidopsis ARGONAUTE1-mediated slicing, transgene-induced RNA silencing, and DNA methylation by distinct domains of the Cucumber mosaic virus 2b protein. *Plant Cell* **24**, 259–274 (2012).
- Zhu, C. et al. A fungal effector suppresses the nuclear export of AGO1-miRNA complex to promote infection in plants. *Proc. Natl Acad. Sci. USA* **119**, e2114583119 (2022).
- Qiao, Y. et al. Oomycete pathogens encode RNA silencing suppressors. *Nat. Genet.* **45**, 330–333 (2013).
- Hou, Y. et al. A phytophthora effector suppresses trans-kingdom RNAi to promote disease susceptibility. *Cell Host Microbe* **25**, 153–165 e155 (2019).
- Qiao, Y., Shi, J., Zhai, Y., Hou, Y. & Ma, W. Phytophthora effector targets a novel component of small RNA pathway in plants to promote infection. *Proc. Natl Acad. Sci. USA* **112**, 5850–5855 (2015).

18. Lichinchi, G. et al. Dynamics of the human and viral m(6)A RNA methylomes during HIV-1 infection of T cells. *Nat. Microbiol.* **1**, 16011 (2016).
19. Kennedy, E. M. et al. Posttranscriptional m(6)A Editing of HIV-1 mRNAs Enhances Viral Gene Expression. *Cell Host Microbe* **19**, 675–685 (2016).
20. Tirumuru, N. et al. N(6)-methyladenosine of HIV-1 RNA regulates viral infection and HIV-1 Gag protein expression. *eLife* **5**, <https://doi.org/10.7554/eLife.15528> (2016).
21. Gokhale, N. S. et al. N6-Methyladenosine in Flaviviridae Viral RNA Genomes Regulates Infection. *Cell Host Microbe* **20**, 654–665 (2016).
22. Martinez-Perez, M. et al. Plant YTHDF proteins are direct effectors of antiviral immunity against an N6-methyladenosine-containing RNA virus. *EMBO J.* **42**, e113378 (2023).
23. Zhang, K. et al. The dynamics of N(6)-methyladenine RNA modification in interactions between rice and plant viruses. *Genome Biol.* **22**, 189 (2021).
24. He, H., Ge, L., Li, Z., Zhou, X. & Li, F. Pepino mosaic virus antagonizes plant m(6)A modification by promoting the autophagic degradation of the m(6)A writer HAKAI. *aBIOTECH* **4**, 83–96 (2023).
25. Zhang, T. et al. N6-methyladenosine RNA modification promotes viral genomic RNA stability and infection. *Nat. Commun.* **13**, 6576 (2022).
26. Hou, W. N., Duan, C. G., Fang, R. X., Zhou, X. Y. & Guo, H. S. Satellite RNA reduces expression of the 2b suppressor protein resulting in the attenuation of symptoms caused by Cucumber mosaic virus infection. *Mol. plant Pathol.* **12**, 595–605 (2011).
27. Deleris, A. et al. Hierarchical action and inhibition of plant Dicer-like proteins in antiviral defense. *Science* **313**, 68–71 (2006).
28. Diaz-Pendon, J. A., Li, F., Li, W. X. & Ding, S. W. Suppression of antiviral silencing by cucumber mosaic virus 2b protein in Arabidopsis is associated with drastically reduced accumulation of three classes of viral small interfering RNAs. *Plant Cell* **19**, 2053–2063 (2007).
29. Wang, X. B. et al. RNAi-mediated viral immunity requires amplification of virus-derived siRNAs in Arabidopsis thaliana. *Proc. Natl Acad. Sci. USA* **107**, 484–489 (2010).
30. Garcia-Ruiz, H. et al. Arabidopsis RNA-dependent RNA polymerases and dicer-like proteins in antiviral defense and small interfering RNA biogenesis during Turnip Mosaic Virus infection. *Plant Cell* **22**, 481–496 (2010).
31. Fang, Y. Y. et al. CMV2b-AGO interaction is required for the suppression of rdr-dependent antiviral silencing in arabidopsis. *Front. Microbiol.* **7**, 1329 (2016).
32. Baquero-Perez, B. et al. N(6)-methyladenosine modification is not a general trait of viral RNA genomes. *Nat. Commun.* **15**, 1964 (2024).
33. Wang, C. et al. Myricetin derivative LP11 targets cucumber mosaic virus 2b protein to achieve in vivo antiviral activity in plants. *J. Agric. food Chem.* **70**, 15360–15370 (2022).
34. Shen, L. Functional interdependence of N6-methyladenosine methyltransferase complex subunits in Arabidopsis. *Plant Cell* **35**, 1901–1916 (2023).
35. Tong, J. et al. ALBA proteins confer thermotolerance through stabilizing HSF messenger RNAs in cytoplasmic granules. *Nat. Plants* **8**, 778–791 (2022).
36. Cai, Z. et al. The m6A reader ECT8 is an abiotic stress sensor that accelerates mRNA decay in Arabidopsis. *The Plant cell*, <https://doi.org/10.1093/plcell/kcae149> (2024).
37. Wu, X. et al. N(6)-methyladenosine-mediated feedback regulation of abscisic acid perception via phase-separated ECT8 condensates in Arabidopsis. *Nat. Plants* **10**, 469–482 (2024).
38. Crawshaw, S., Watt, L. G., Murphy, A. M. & Carr, J. P. Strain-specific differences in the interactions of the cucumber mosaic virus 2b protein with the viral 1a and host Argonaute 1 proteins. *J. Virol.* **98**, e0099324 (2024).
39. Kim, H., Shimura, H., Sueda, K. & Masuta, C. Importin/exportin-mediated nucleocytoplasmic shuttling of cucumber mosaic virus 2b protein is required for 2b's efficient suppression of RNA silencing. *PLoS Pathog.* **18**, e1010267 (2022).
40. Du, Z. et al. Nuclear-cytoplasmic partitioning of cucumber mosaic virus protein 2b determines the balance between its roles as a virulence determinant and an RNA-silencing suppressor. *J. Virol.* **88**, 5228–5241 (2014).
41. Wei, L. H. et al. The m(6)A Reader ECT2 controls trichome morphology by affecting mrna stability in arabidopsis. *Plant Cell* **30**, 968–985 (2018).
42. Truman, W. et al. The calmodulin-binding protein60 family includes both negative and positive regulators of plant immunity. *Plant Physiol.* **163**, 1741–1751 (2013).
43. Ding, Y. et al. Opposite roles of salicylic acid receptors NPR1 and NPR3/NPR4 in transcriptional regulation of plant immunity. *Cell* **173**, 1454–1467 e1415 (2018).
44. Zhang, T. et al. Cotton plants export microRNAs to inhibit virulence gene expression in a fungal pathogen. *Nat. plants* **2**, 16153 (2016).
45. Brocard, M., Ruggieri, A. & Locker, N. m6A RNA methylation, a new hallmark in virus-host interactions. *J. Gen. Virol.* **98**, 2207–2214 (2017).
46. Yue, J. et al. AlkB RNA demethylase homologues and N(6) -methyladenosine are involved in Potyvirus infection. *Mol. plant Pathol.* **23**, 1555–1564 (2022).
47. He, H. et al. m(6)A modification of plant virus enables host recognition by NMD factors in plants. *Sci. China Life Sci.* **67**, 161–174 (2024).
48. Zhang, X. et al. Cucumber mosaic virus-encoded 2b suppressor inhibits Arabidopsis Argonaute1 cleavage activity to counter plant defense. *Genes Dev.* **20**, 3255–3268 (2006).
49. Guo, Z. et al. Lipid flippases promote antiviral silencing and the biogenesis of viral and host siRNAs in Arabidopsis. *Proc. Natl Acad. Sci. USA* **114**, 1377–1382 (2017).
50. Guo, Z. et al. Identification of a new host factor required for antiviral RNAi and Amplification of Viral siRNAs. *Plant Physiol.* **176**, 1587–1597 (2018).
51. Wang, X. B. et al. The 21-nucleotide, but not 22-nucleotide, viral secondary small interfering RNAs direct potent antiviral defense by two cooperative argonautes in Arabidopsis thaliana. *Plant Cell* **23**, 1625–1638 (2011).
52. Bhat, S. S. et al. mRNA adenosine methylase (MTA) deposits m(6)A on pri-miRNAs to modulate miRNA biogenesis in Arabidopsis thaliana. *Proc. Natl Acad. Sci. USA* **117**, 21785–21795 (2020).
53. Bai, H. et al. The METHYLTRANSFERASE B-SERRATE interaction mediates the reciprocal regulation of microRNA biogenesis and RNA m(6)A modification. *J. Integr. plant Biol.* <https://doi.org/10.1111/jipb.13770> (2024).
54. Bodi, Z. et al. Adenosine methylation in arabidopsis mrna is associated with the 3' end and reduced levels cause developmental defects. *Front. Plant Sci.* **3**, 48 (2012).
55. Guo, Z., Wang, X. B., Li, W. X. & Ding, S. W. A sensitized genetic screen to identify novel components and regulators of the host antiviral rna interference pathway. *Methods Mol. Biol.* **2028**, 215–229 (2019).
56. Yan, W. et al. Discovery of aphid-transmitted Rice tiller inhibition virus from native plants through metagenomic sequencing. *PLoS Pathog.* **19**, e1011238 (2023).
57. Pall, G. S. & Hamilton, A. J. Improved northern blot method for enhanced detection of small RNA. *Nat. Protoc.* **3**, 1077–1084 (2008).
58. Xie, S. S. et al. MJ28 guides sequence-specific targeting of ATX1/2-containing COMPASS-like complex in Arabidopsis. *Cell Rep.* **42**, 112163 (2023).

59. Wang, X. et al. N6-methyladenosine-dependent regulation of messenger RNA stability. *Nature* **505**, 117–120 (2014).
60. Bai, H. et al. The methyltransferase b-serrate interaction mediates the reciprocal regulation of microRNA biogenesis and RNA m(6)A modification. *J. Integr. plant Biol.* **66**, 2613–2631 (2024).
61. Bolger, A. M., Lohse, M. & Usadel, B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* **30**, 2114–2120 (2014).
62. Li, H. & Durbin, R. Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* **25**, 1754–1760 (2009).
63. Dominissini, D., Moshitch-Moshkovitz, S., Salmon-Divon, M., Amariglio, N. & Rechavi, G. Transcriptome-wide mapping of N(6)-methyladenosine by m(6)A-seq based on immunocapturing and massively parallel sequencing. *Nat. Protoc.* **8**, 176–189 (2013).
64. Meng, J. et al. A protocol for RNA methylation differential analysis with MeRIP-Seq data and exomePeak R/Bioconductor package. *Methods* **69**, 274–281 (2014).
65. Cui, X. et al. MeTDiff: A Novel Differential RNA Methylation Analysis for MeRIP-Seq Data. *IEEE/ACM Trans. computational Biol. Bioinforma.* **15**, 526–534 (2018).
66. Zhang, Y. et al. Model-based analysis of ChIP-Seq (MACS). *Genome Biol.* **9**, R137 (2008).
67. Klopfenstein, D. V. et al. GOATOOLS: A Python library for Gene Ontology analyses. *Sci. Rep.* **8**, 10872 (2018).
68. Dobin, A. et al. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* **29**, 15–21 (2013).
69. Li, B. & Dewey, C. N. RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. *BMC Bioinforma.* **12**, 323 (2011).
70. Robinson, M. D., McCarthy, D. J. & Smyth, G. K. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* **26**, 139–140 (2010).
71. Shen, W., Le, S., Li, Y. & Hu, F. SeqKit: a cross-platform and ultrafast toolkit for FASTA/Q file manipulation. *PloS one* **11**, e0163962 (2016).
72. Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* **34**, 3094–3100 (2018).
73. Li, H. et al. The sequence alignment/map format and SAMtools. *Bioinformatics* **25**, 2078–2079 (2009).

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Author contributions

C.G.D. and J.H.L. designed this study. J.H.L. and Y.X.L. conducted the experiments. Y.L. performed the computational analysis of the MeRIP-seq, DRS-seq, and RNA-seq data. Z.L., Z.Z. and C.G.D. analyzed the data. C.G.D. wrote the manuscript. All authors have read, discussed, and contributed to the manuscript. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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